Research on the Design of Optimal Polarization Modes for Generalized Compact Polarimetry SAR Target Classification

Guo Song, Yunkai Deng, Heng Zhang, Xiuqing Liu, Nan Wang, Yuanbo Jiao, Wentao Hou, and Xingjie Zhao

***Abstract*— This article proposes a generalized compact polarimetry (GCP) mode along with two optimal polarization mode selection parameters to address the challenges of polarization mode selection in classification tasks across diverse scenarios. Theoretically, we conduct an in-depth analysis of the differences between circular and linear transmit polarizations, demonstrating their fundamental equivalence in terms of information content. For the first time, we propose that different classification tasks require different optimal polarization modes, and the optimal transmit polarization mode may lie in the elliptic polarization domain of synthetic aperture radar (SAR) systems rather than traditional circular compact polarimetric (CP) or linear dual-polarization (DP) modes. The proposed approach is validated using full-polarimetric SAR data from San Francisco and Hainan, showing that the optimal elliptical polarization mode achieves classification accuracies that are 2% to 42% higher than those of traditional CP or DP modes for certain categories, and performs comparably to full polarization. This improvement in accuracy stems from the interaction between the transmit polarization and the target scene, rather than advancements in classification algorithms. Using the two proposed parameters, the overall and category-specific classification performance of GCP modes can be effectively evaluated, enabling the identification of the optimal polarization mode for a given task. These findings provide significant insight into the design of future polarimetric SAR systems and offer new perspectives and directions for mission planning and mode selection for on-orbit satellites.**

***Index Terms*— Elliptical polarization generalized compact polarimetry (GCP), optimal polarization modes, synthetic aperture radar (SAR) target classification, Wishart distance.**

I. Introduction

SYNTHETIC aperture radar (SAR) has been extensively researched and applied due to its all-weather, all-day operational capabilities [1]. Since the introduction of the first polarimetric radar concept in the early 1950s, significant advancements have been made in radar polarimetry. These advancements have enabled wide-ranging applications including natural surface segmentation and classification [2], [3], forest parameter estimation [4], [5], roughness state classification [6], soil moisture mapping [7], lava flow characterization [8], ice and snow detection and inversion [9], [10], [11], topographic mapping [12], urban classification [13], and target detection and analysis [14].

SAR has evolved from the initial single-polarization mode to dual-polarization (DP), quad-polarimetric (QP), and compact polarimetric (CP) modes [15], [16]. A single-polarization SAR system transmits and receives a single polarized electromagnetic wave, capturing only the backscatter intensity information of the target. DP and CP SAR systems transmit a single polarized electromagnetic wave and receive two orthogonally polarized waves, thereby obtaining more polarimetric information than single-polarization SAR systems. Compared to single-polarization and DP SAR, QP SAR systems operate in a "transmit alternately, receive simultaneously" mode, alternately transmitting two orthogonally polarized electromagnetic waves and receiving two orthogonally polarized waves simultaneously. This allows the acquisition of complete polarimetric information on the target, significantly enhancing the SAR system's capability for accurate classification and offering stronger application potential.

However, obtaining complete polarimetric information with fully polarimetric SAR comes with costs such as narrower swath width and increased system design complexity, which include the following.

1) Fully polarimetric SAR alternately transmits H and V polarized waves, requiring the pulse repetition frequency (PRF) to be doubled compared to single-polarization and DP SAR. This theoretically halves the swath width in the range direction for fully polarimetric SAR [17].
2) In most scenes, the cross-polarization (HV/VH) channel echo power is 6–10 dB lower than the co-polarization (HH/VV) channel echo power. To compensate for this difference, the fully polarimetric receiving system needs to employ channel gain control and complex internal calibration loops, increasing system design complexity.
3) The narrower range swath and the imbalance in echo power between cross-polarization and co-polarization

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channels degrade the range ambiguity performance of the fully polarimetric SAR cross-polarization channel. To meet range ambiguity performance requirements, further restrictions on the SAR incidence angle range and swath width are necessary [18].

To address the aforementioned drawbacks of fully polarimetric SAR, such as narrower swath width and increased system complexity, Souyris et al. [19] first introduced the concept of CP SAR in 2005. Currently, more and more spaceborne SAR systems are equipped with CP modes [20], [21], [22], [23], [24], [25].

Over the past decade, CP SAR has garnered significant attention in the field of polarimetric SAR. There has been extensive research on CP SAR decomposition methods and classification applications. As research into CP SAR has deepened, many researchers believe that it overcomes the disadvantages of QP SAR while essentially achieving its polarimetric capabilities [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36]. Compared to QP and CP SAR systems, researchers often either directly overlook DP SAR in favor of studying CP SAR or assume that DP SAR performs worse than CP SAR without providing sufficiently convincing reasons. The following research status will demonstrate this to readers.

Some classification application studies have compared the classification performance of CP, DP, and QP SAR as follows: Ainsworth et al. [37] compared the Wishart classification accuracy of QP, DP, and CP SAR. The experimental results showed that QP SAR had the highest classification accuracy, while CP and DP SAR modes could often match the classification accuracy of QP SAR. Ainsworth pointed out that the transmitted polarization determines the information obtained from SAR images, and the scattering characteristics of the imaging scene determine the optimal choice of transmitted polarization. CP modes generally perform better in image classification than DP modes, but this is not always the case. For instance, features such as fences around and on airport runways can be observed with DP modes but not with CP modes [37]. Wu et al. [38] used QP data from RADARSAT-2 to monitor rice growth and noted that for retrieving rice parameters such as biomass or height, HV or VH data were more suitable than HH and VV data. Among all four polarization datasets, VV polarization had the lowest correlation with rice age and growth parameters. The HH/VV ratio image showed the greatest distinction between rice crops, bananas, forests, and rivers, indicating that there is an optimal transmitted polarization for rice growth monitoring applications [38]. Li et al. [39] identified HH/HV as the best DP combination for rice classification. Xie et al. [40] analyzed and compared the decompositions of CP and DP modes by considering the differences in polarimetric characteristics between targets and their surroundings. The decomposition results were used for maritime applications such as oil spill monitoring, coastline extraction, and ship detection, but they did not specify which polarization mode was superior [40]. Brisco et al. [41] pointed out that for rice and wetland mapping classification applications, circular polarization coherence performed best for both rice and wetland classification. While CP and QP data achieved sufficient classification accuracy for practical applications, CP data performed better than DP data but not as well as QP data. However, due to the wider swath coverage, this mode might offer the best option for rice and wetland mapping applications [41]. Larrañaga and Álvarez-Mozos [42] used a random forest classification algorithm for object-based supervised classification in multitemporal crop classification. The overall accuracy of dual-polarization data is the same as that of full-polarization data. Yin et al. [43] proposed the αBCP/$\Delta\alpha$BCP plane for general CP classification, the only classification method applicable to all CP modes to date. Yin noted that the performance of the αBCP/$\Delta\alpha$BCP plane depends on the target characteristics captured by the CP SAR mode. Yin analyzed the effects of orientation and helicity angles on target characteristics under different CP modes. The results showed that the circular CP mode is least affected by target orientation and helicity when interpreting target scattering mechanisms and provides results most similar to QP data. However, this does not mean that the circular CP mode is suitable for all types of observation scenes; the linear CP mode performed better in distinguishing forest and urban categories. The optimal CP mode for different applications and observation scenes still requires careful study [43]. Hou et al. [44] pointed out that previous comparisons of classification performance between CP and DP modes were not rigorous and did not fairly compare the polarimetric information contained in the data. They proposed a unified framework to fairly compare the classification performance of different polarimetric systems based on Wishart mixture model (WMM). They suggested that under current mainstream views, the CP mode should not be considered the "golden configuration." CP modes have less classification potential in urban areas compared to DP modes, but the reasons for this were not thoroughly explored [44].

Current research on CP SAR classification still has several limitations. First, many papers claim that CP classification performs better than DP in classification applications. This is an inaccurate conclusion, as seen from the aforementioned studies, where CP and DP modes each have their respective strengths and weaknesses in different application scenarios. Second, there is a lack of in-depth systematic research on the optimal polarization mode types for classification applications, especially generalized CP classification studies that include elliptical CP modes. Moreover, there is a lack of methods designed to determine the optimal generalized CP mode for specific classification applications.

The main contributions of this article are as follows. This article proposes optimal polarization mode selection parameters for target classification to measure the classification performance of generalized compact polarimetry (GCP) modes, which include CP, DP, and elliptical polarization modes. These parameters are used to plan the optimal polarization mode for classification tasks. This article analyzes and compares the similarities and differences between CP and DP modes, theoretically proving that their polarimetric information content is essentially consistent and analyzing the reasons for their differences in classification performance. The

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core viewpoint of this article is that no specific CP mode necessarily outperforms other modes in classification performance, and different classification tasks have corresponding optimal polarization modes. This viewpoint contrasts with the current mainstream belief that CP SAR's overall performance surpasses that of traditional DP SAR. The related conclusion contributes to understanding the performance of generalized CP SAR and provides guidance for the design of future polarimetric SAR systems.

The structure of this article is as follows. Section II introduces the GCP mode, compares the differences between single-transmit dual-receive systems, and explains that the information content across different polarization modes is essentially the same. Section III proposes the optimal polarization mode selection parameters. Section IV compares the classification results of different polarization modes, analyzes the classification performance of GCP SAR, and validates the effectiveness of the proposed optimal GCP SAR design method through experiments. Further analysis is presented in Section V. Section VI concludes this article.

II. GCP Mode

This section proposes referring to all single-transmit dual-receive polarization modes as GCP modes. It analyzes the similarities and differences among various single-transmit dual-receive systems, demonstrating that the information content of the transmitted electromagnetic waves is theoretically consistent across these systems. The differences in classification performance among the systems arise from the varying interactions between the transmitted electromagnetic waves and the ground objects. Each GCP mode has its own area where it achieves the best classification results.

A. Proposal of GCP Mode

Essentially, GCP SAR systems are single-transmit dual-receive SAR systems. We propose referring to all single-transmit dual-receive polarization modes, including DP and compact polarization modes, as GCP modes. CP SAR can still be considered a type of DP mode, i.e., a single-transmit dual-receive mode.

In principle, there is no fundamental difference between CP SAR and DP SAR. In CP SAR, the $\pi/4$ mode transmits 45° linearly polarized electromagnetic waves, while DP SAR transmits H and V linearly polarized electromagnetic waves. Both transmit linearly polarized electromagnetic waves; the only difference lies in the transmission angle, with both systems receiving H and V linearly polarized signals. For the dual circular polarization (DCP) and circular polarization transmitting linear polarization receiving (CTLR) modes of CP SAR, although it may appear that left-hand or right-hand circular polarization is synthesized from H and V linear polarization, in reality, CP SAR selects only one of the circular polarizations for transmission, not transmitting complete polarization signals. This is different from QP SAR, which alternately transmits H and V polarization signals.

The circular polarization signals transmitted in CP SAR can be synthesized from the H and V linear polarization basis, similar to how linearly polarized signals H or V can be synthesized from the circular polarization basis R and L in DP SAR. Therefore, we believe there is no fundamental difference in the operational mode between CP SAR and DP SAR.

Polarization modes include not only circular and linear polarization but also elliptical polarization. All polarization states can be described using the two parameters of the polarization ellipse: the ellipticity angle and the orientation angle. When the ellipticity angle is 0°, and the orientation angle is 0°, 45°, or 90°, the polarization states are H, 45°, and V linear polarization, respectively. When the ellipticity angle is 45°, the polarization state is circular polarization, and in other cases, the polarization state is typically elliptical polarization.

There is no fundamental difference between DP and CP modes. They are, respectively, "DP" and "CP" in the circular and linear polarization bases. Section II-B will introduce this concept. Considering that both DP and CP modes are single-transmit dual-receive modes and that the transmitted electromagnetic wave in single-transmit dual-receive polarization SAR modes can be linearly, circularly, or elliptically polarized, and since elliptical polarization encompasses countless polarization modes with linear and circular polarization being special cases, it follows that DP and CP modes are special cases among the numerous single-transmit dual-receive polarization modes. Therefore, we propose to refer to all single-transmit dual-receive polarization modes as GCP modes, which include single-transmit dual-receive modes that transmit elliptically polarized waves.

The current polarimetric SAR systems can achieve both linear and circular polarizations. The hardware differences required to enable antennas capable of generating elliptically polarized waves can be discussed in two parts.

1) *No Hardware Modification:* SAR systems can generate certain elliptical polarizations without hardware changes, which are identical to linear and circular polarization antennas. This is detailed in Appendix A.
2) *With Hardware Modification:* SAR systems require specific hardware modifications to achieve elliptical polarization with both adjustable ellipticity angle and orientation angle. These modifications can be realized using two approaches: a) *sparse phased arrays:* By adjusting the amplitude between horizontal and vertical channels, the orientation angle of the elliptical polarization can be controlled and b) *multipath synthesis design at the final-stage power amplifier of the radar transmission channel:* This design allows selective activation of certain branches to meet the specific feeding amplitude requirements for certain elliptical polarizations, particularly when lower feeding amplitudes are required for specific modes of elliptical polarization. By combining these two methods, SAR systems can better achieve the generation of elliptical polarization waves.

The above method achieves elliptical polarization through the synthesis of horizontal and vertical channels, which is how most SAR systems currently implement circular polarization transmission. Theoretically, elliptical polarization can also be achieved without synthesis, by directly transmitting

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elliptical polarization through the antenna, which requires further investigation.

B. Comparison of Transmission and Reception in Single-Transmit Dual-Receive Polarization Mode

Polarized electromagnetic waves can be described using a pair of mutually orthogonal polarization bases [45]. We typically use linear polarization bases h and v or circular polarization bases r and l. For now, let us consider circular polarization and linear polarization as special cases of elliptical polarization, which are also the most common polarizations used in CP and DP transmission and reception. In this article, we use lowercase letters h, v, r, and l to represent polarization bases, uppercase letters H, V, R, and L to represent corresponding polarization modes, and S_{PQ} to represent the scattering vector corresponding to Q polarization for transmission and P polarization for reception.

In the linear polarization basis $[h, v]$ and the circular polarization basis $[l, r]$, horizontal polarization H, vertical polarization V, left-hand circular polarization L, and right-hand circular polarization R can be expressed as

$$\begin{aligned}
\mathrm{H} &= (1,0)\begin{bmatrix} h \\ v \end{bmatrix} = \frac{1}{\sqrt{2}}(1,-j)\begin{bmatrix} l \\ r \end{bmatrix} \\
\mathrm{V} &= (0,1)\begin{bmatrix} h \\ v \end{bmatrix} = \frac{1}{\sqrt{2}}(-j,1)\begin{bmatrix} l \\ r \end{bmatrix} \\
\mathrm{L} &= \frac{1}{\sqrt{2}}(1,j)\begin{bmatrix} h \\ v \end{bmatrix} = (1,0)\begin{bmatrix} l \\ r \end{bmatrix} \\
\mathrm{R} &= \frac{1}{\sqrt{2}}(j,1)\begin{bmatrix} h \\ v \end{bmatrix} = (0,1)\begin{bmatrix} l \\ r \end{bmatrix}. \qquad (1)
\end{aligned}$$

It can be observed that each polarization basis can be represented in terms of the others.

For these two pairs of polarization bases, four types of transmit–receive polarization combinations can be obtained, respectively. The linear polarization basis provides $[HH, HV, VH, VV]^{\mathrm{T}}$, where "HV" indicates transmitting vertical polarization and receiving horizontal polarization. In this notation, the first letter denotes the receiving polarization and the second letter denotes the transmitting polarization. This set of combinations represents all possible combinations of transmitting and receiving horizontal (H) and vertical (V) polarizations.

Similarly, the circular polarization basis yields $[LL, LR, RL, RR]^{\mathrm{T}}$. For the scenario where circular polarization (R or L) is transmitted and linear polarization (H or V) is received, the combinations of echo signals are $[HR, HL, VR, VL]^{\mathrm{T}}$.

From this, it can be deduced that these polarization combination echo signals can be converted among each other

$$\begin{aligned}
\begin{bmatrix} HH \\ HV \\ VH \\ VV \end{bmatrix} &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\begin{bmatrix} HH \\ HV \\ VH \\ VV \end{bmatrix} \\
&= \frac{1}{2}\begin{pmatrix} 1 & -j & -j & -1 \\ -j & 1 & -1 & -j \\ -j & -1 & 1 & -j \\ -1 & -j & -j & 1 \end{pmatrix}\begin{bmatrix} LL \\ LR \\ RL \\ RR \end{bmatrix} \\
\begin{bmatrix} LL \\ LR \\ RL \\ RR \end{bmatrix} &= \frac{1}{2}\begin{pmatrix} 1 & j & j & -1 \\ j & 1 & -1 & j \\ j & -1 & 1 & j \\ -1 & j & j & 1 \end{pmatrix}\begin{bmatrix} HH \\ HV \\ VH \\ VV \end{bmatrix} \\
&= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\begin{bmatrix} LL \\ LR \\ RL \\ RR \end{bmatrix} \\
\begin{bmatrix} HR \\ HL \\ VR \\ VL \end{bmatrix} &= \frac{\sqrt{2}}{2}\begin{pmatrix} j & 1 & 0 & 0 \\ 1 & j & 0 & 0 \\ 0 & 0 & j & 1 \\ 0 & 0 & 1 & j \end{pmatrix}\begin{bmatrix} HH \\ HV \\ VH \\ VV \end{bmatrix} \\
&= \frac{\sqrt{2}}{2}\begin{pmatrix} 0 & 1 & 0 & -j \\ 1 & 0 & -j & 0 \\ 0 & -j & 0 & 1 \\ -j & 0 & 1 & 0 \end{pmatrix}\begin{bmatrix} LL \\ LR \\ RL \\ RR \end{bmatrix}. \qquad (2)
\end{aligned}$$

It can be observed that there exists a conversion relationship between the echo signal combinations of linear polarization and circular polarization. Once the target response in one set of polarization bases is obtained, the polarization response in any other bases can also be obtained without additional measurements, using straightforward mathematical transformations.

Taking the full polarization SAR system as an example, the system receives four polarization combination echoes, and the measured Sinclair matrix is

$$S_{QP} = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix}. \qquad (3)$$

To conveniently see the relationship between each polarization SAR working mode and each polarization combination echo, we rewrite (3) into the form of a column vector as follows:

$$\begin{aligned}
S_{QP} &= \begin{bmatrix} S_{HH} \\ S_{HV} \\ S_{VH} \\ S_{VV} \end{bmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\begin{bmatrix} HH \\ HV \\ VH \\ VV \end{bmatrix} \\
&= \frac{1}{2}\begin{pmatrix} 1 & -j & -j & -1 \\ -j & 1 & -1 & -j \\ -j & -1 & 1 & -j \\ -1 & -j & -j & 1 \end{pmatrix}\begin{bmatrix} LL \\ LR \\ RL \\ RR \end{bmatrix}. \qquad (4)
\end{aligned}$$

The DP SAR system transmits horizontal linear polarization waves or vertical linear polarization waves, and receives horizontal and vertical polarization waves at the same time. Taking the DP SAR system that transmits horizontal linear polarization waves as an example, the system receives two polarization combination echoes, and the measured scattering vector is

$$\begin{aligned}
S_{\mathrm{DP}} &= \begin{bmatrix} S_{HH} \\ S_{VH} \end{bmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}\begin{bmatrix} HH \\ HV \\ VH \\ VV \end{bmatrix} \\
&= \frac{1}{2}\begin{pmatrix} 1 & -j & -j & -1 \\ -j & -1 & 1 & -j \end{pmatrix}\begin{bmatrix} LL \\ LR \\ RL \\ RR \end{bmatrix}. \qquad (5)
\end{aligned}$$

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TABLE I
COMPARISON OF SCATTERING VECTORS FOR SINGLE-TRANSMIT DUAL-RECEIVE POLARIZATION MODE TRANSMISSION AND RECEPTION

Polarization mode	Transmit polarization	Receive polarization	Linear polarization basis	circularly polarized basis
DCP	L or R	L and R	$\frac{1}{2}\begin{bmatrix}-1 & j & j & 1\\ j & 1 & -1 & j\end{bmatrix}$	$\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix}$
CTLR	L or R	H and V	$\frac{\sqrt{2}}{2}\begin{bmatrix}j & 1 & 0 & 0\\ 0 & 0 & j & 1\end{bmatrix}$	$\frac{\sqrt{2}}{2}\begin{bmatrix}-j & 0 & 1 & 0\\ 1 & 0 & -j & 0\end{bmatrix}$
DP	H or V	H and V	$\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix}$	$\frac{1}{2}\begin{bmatrix}-1 & -j & -j & 1\\ -j & 1 & -1 & -j\end{bmatrix}$

For comparison, the transmission of R or H is taken as an example here.

The scattering vectors for DCP mode and CTLR mode refer to Appendix B

$$S_{\mathrm{DCP}} = \begin{bmatrix} S_{RR} \\ S_{LR} \end{bmatrix} = \begin{bmatrix} jS_{HR} + S_{VR} \\ S_{HR} + jS_{VR} \end{bmatrix} = \begin{bmatrix} j & 1 \\ 1 & j \end{bmatrix} \begin{bmatrix} S_{HR} \\ S_{VR} \end{bmatrix} = \begin{bmatrix} j & 1 \\ 1 & j \end{bmatrix} S_{\mathrm{CTLR}}. \tag{6}$$

In the context of polarimetric SAR systems, the DCP mode can be achieved through receiving polarization combinations similar to those in the CTLR mode, as shown in (6). Therefore, the main difference between DCP and CTLR modes lies in their receiving polarization bases. These modes can be transformed into each other through simple matrix operations, and there is no difference in the amount of information they convey.

As shown in Table I, DCP and DP modes exhibit significant duality. Specifically, under the linear polarization basis, the DCP mode resembles the DP mode under the circular polarization basis: the scattering vector for the DCP mode under the linear polarization basis is represented as $(1/2)\begin{bmatrix}-1 & j & j & 1\\ j & 1 & -1 & j\end{bmatrix}$, while the scattering vector for the DP mode under the circular polarization basis is represented as $(1/2)\begin{bmatrix}-1 & -j & -j & 1\\ -j & 1 & -1 & -j\end{bmatrix}$. It can be observed that DP differs from DCP only by a negative sign in front of j, reflecting a difference in direction only. Conversely, under the circular polarization basis, the DCP mode shows the exact same form as the DP mode under the linear polarization basis. This indicates that the difference between them primarily stems from the choice of polarization bases, resulting in variations in the representation of the scattering vector. From the perspective of information content, the scattering vectors of both modes are equivalent

$$S_{\mathrm{DCP}} = \begin{bmatrix} S_{HR} \\ S_{VR} \end{bmatrix} = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \begin{bmatrix} 1 \\ -j \end{bmatrix} \tag{7}$$

$$S_{\mathrm{DP}} = \begin{bmatrix} S_{HH} \\ S_{VH} \end{bmatrix} = \begin{bmatrix} S_{LL} & S_{LR} \\ S_{RL} & S_{RR} \end{bmatrix} \begin{bmatrix} 1 \\ -j \end{bmatrix}. \tag{8}$$

Equations (7) and (8) show the relationship between the scattering vector of DCP mode emitting R and linear full polarization, as well as the relationship between DP mode emitting H and circular full polarization. It can be seen that they only differ in polarization basis. The DCP mode, which transmits circularly polarized waves, can be regarded as a compact polarization mode under the linear polarization basis. Conversely, the DP mode, which transmits linearly polarized waves, can be seen as a "compact polarization" mode under the circular polarization basis.

C. Comparison of Single-Transmit Dual-Receive Polarization Mode on Backscattering of Typical Targets

Unlike the comprehensive polarization information and optimal performance obtained from QP modes, single-transmit dual-receive polarization modes fail to capture complete polarization information. Therefore, these modes cannot be converted into each other. Within the same polarization basis, there exists a difference in the amount of information obtained between DP and compact polarization modes. This leads to our key observation: the GCP mode shows no fundamental difference in the information content of the transmitted electromagnetic waves, and their information content is equivalent; the specific differences in information content between different types of GCP modes such as DP and compact polarization are caused by the polarization scattering characteristics of the objects being observed. The echo signal results from the interaction between emitted electromagnetic waves and the observed object. The polarization mode of the emitted electromagnetic waves, the geometric structure of the observed object, and its dielectric properties all influence the amount of information contained in the echo signal.

Therefore, we propose that due to specific differences in objects, different GCP modes each have their advantages in classifying application domains. Next, we will compare the impact differences of different polarization modes on the scattering matrices of typical targets by analyzing compact polarization modes and dual polarization modes.

1) Scattering Matrices of Typical Targets Under Different Transmission Bases: For the CP and DP SAR working modes, the main emission polarization types are linear polarization waves and circular polarization waves. The scattering matrices of several typical targets under different emission bases are shown in Table II. For details, please refer to Appendix C

In the aforementioned scattering scenarios, we observe the following distinctions: when transmitting horizontally polarized waves, it is difficult to distinguish between horizontal dipole scattering, surface scattering, and double-bounce scattering. Similarly, when transmitting vertically polarized waves, distinguishing between vertical dipole scattering and

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TABLE II
SCATTERING MATRICES OF TYPICAL TARGETS UNDER DIFFERENT TRANSMISSION BASES

Transmit polarization	H	V	R	L
Surface scattering	$[1 \quad 0]$	$[0 \quad 1]$	$[1 \quad -j]$	$[1 \quad j]$
double-bounce scattering	$[1 \quad 0]$	$[0 \quad -1]$	$[1 \quad j]$	$[1 \quad -j]$
Rotation dihedral reflector	$[\cos 2\theta \quad \sin 2\theta]$	$[\sin 2\theta \quad -\cos 2\theta]$	$[1 \quad j]$	$[1 \quad -j]$
22.5°dihedral reflector	$[1 \quad 1]$	$[1 \quad -1]$	$[1 \quad j]$	$[1 \quad -j]$
-22.5°dihedral reflector	$[1 \quad -1]$	$[1 \quad 1]$	$[1 \quad j]$	$[1 \quad -j]$
45°dihedral reflector	$[0 \quad 1]$	$[1 \quad 0]$	$[1 \quad j]$	$[1 \quad -j]$
Horizontal dipole	$[1 \quad 0]$	$[0 \quad 0]$	$[1 \quad 0]$	$[1 \quad 0]$
Vertical dipole	$[0 \quad 0]$	$[0 \quad 1]$	$[0 \quad 1]$	$[0 \quad 1]$
Right-hand helix	$[1 \quad -j]$	$[1 \quad -j]$	$[0 \quad 0]$	$[1 \quad -j]$
Left-hand helix	$[1 \quad j]$	$[1 \quad j]$	$[1 \quad j]$	$[0 \quad 0]$

The scattering matrices are represented in the [h v] polarization basis. The yellow highlighting in the same column indicates that different typical targets have the same scattering matrix under that particular transmission polarization.

surface scattering is challenging. Furthermore, when transmitting right-hand circularly polarized waves, it is not effective to differentiate between double-bounce scattering, rotation dihedral reflector, and left-hand helix. Conversely, left-hand circularly polarized waves make it difficult to distinguish between double-bounce scattering, rotation dihedral reflectors, and right-hand helix. Based on these observations, we propose a central viewpoint: due to the varying discrimination capabilities of different transmitted polarizations toward specific types of scattering, there are performance differences in the recognition and classification of different targets between CP SAR and DP SAR. This principle similarly applies to other general elliptical polarization modes within GCP.

For SAR images, within a resolution unit, the scattering matrix comes from the coherent superposition of backscattering from all scattering centers

$$S = \sum_{k=1}^{N} \alpha_k S_k. \tag{9}$$

In (9), S represents the scattering matrix of a resolution unit, N is the number of scattering centers within the resolution unit, α_k represents the weight of each scattering center, and S_k is the scattering matrix of the kth scattering center. For regions containing a large number of dihedrals at different angles, although the actual situation may not be as ideal as typical targets, circular polarization tends to be more consistent in the scattering matrix for dihedrals at various angles compared to linear polarization, making it less effective at distinguishing between the angles of the dihedrals. The essence of classification is to distinguish whether different resolution cells belong to the same category under the current resolution. Circular polarization is less effective than linear polarization in eliminating the uncertainty of different cell categories

$$H_p(X) = -\sum_{x} p(x) \cdot \log_2 p(x). \tag{10}$$

For classified ground objects, the contained information entropy H_p is calculated using (10), x is the possible selection of land features, and $p(x)$ is the probability corresponding to that selection. The amount of information obtained using polarization mode A (also referred to in some studies as "Information Gain") is defined as the change in information entropy: $I_A = H_p(X) - H_p(X|A)$. $H_p(X|A)$ represents the information entropy of the ground object when using polarization mode A. In the dihedral example with varying angles, suppose the object could be n dihedrals with equal probabilities of $1/n$. Referring to Table II, for the scattering mechanisms of typical targets, when transmitting R, the received signals from dihedrals at all angles are $[1,j]$, making it impossible to rule out any dihedral angle. The object's information entropy before and after using the circular polarization mode remains unchanged, so the information gain is 0 for R in this scenario. However, when transmitting H, suppose the received signal is $[\cos 2\theta \quad \sin 2\theta]$. Ideally, the probability of the object being dihedral at angle θ becomes 1, ruling out other dihedral angles. According to (10), the calculated information entropy becomes 0. After using the H polarization mode, the object's entropy is reduced to 0, so the information gain is $H_p(X)$, which is the maximum possible information gain in this ideal case. Clearly, in this scenario, linear polarization reduces uncertainty more effectively than circular polarization and provides greater information, making it the superior mode for classification. This conclusion is further reflected in the experimental results presented later in the article. Of course, according to Table II, there are scenarios where linear polarization fails to distinguish

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targets, in which case the information content of circular polarization becomes greater.

The difference between CP and DP modes mainly lies in the different transmission bases, which leads to a scenario where, in specific target recognition and classification tasks, one polarization mode may yield more information than the other.

2) Analysis of scattering of Buildings and Vegetation: For natural terrain, such as calm water surfaces or flat ground, the primary scattering mechanism is usually surface scattering. In contrast, man-made structures, like buildings in urban areas, predominantly exhibit multiple scattering due to the numerous dihedrals they contain. The situation in forested and other vegetated areas is more complex. For vegetated areas, as described by Cloude and Pottier [46], we can classify the scattering mechanisms as follows.

1) *Scattering Mechanism (I):* Backscatter from rough surfaces typically occurs in sparsely vegetated areas with gaps that allow radar waves to penetrate to the ground, such as in sparse forests.
2) *Scattering Mechanism (II):* Low-order multiple scattering arises from dihedrals formed by tree trunks growing perpendicular to the ground, resulting in double scattering.
3) *Scattering Mechanism (III):* Random volume backscatter from the impenetrable layer within the canopy, which contains numerous randomly distributed small scatterers such as crowns, leaves, branches, and low vegetation on the ground, mainly resulting in volume scattering.
4) *Scattering Mechanism (IV):* Surface scattering after propagation through a random medium occurs at lower frequencies with stronger penetration capabilities, such as L- and P-band.
5) *Scattering Mechanism (V):* Single scattering from anisotropic structures, where backscatter can be modeled as originating from rough dielectric cylinders or other objects exhibiting polarization anisotropy due to their shape and dielectric properties, such as tree trunks.

Compared to the scattering from water surfaces and urban buildings, scattering in forested areas is more depolarized [47]. The primary scattering mechanism is volume scattering [48], though there is also some surface and double-bounce scattering, in reality, [49]. As Freeman and Durden [50] noted, L-band backscatter from dense forests is mainly composed of volume and double-bounce scattering, with surface scattering being negligible. However, due to variations in vegetation type and density, the backscatter characteristics of specific vegetation need to be analyzed in detail. Even the seasonal changes in the same crop can affect its backscatter properties, making the selection of the optimal polarization mode particularly important.

3) Analysis of Buildings and Vegetation Using DP and CP Mode: In urban areas, as mentioned earlier, the main scattering mechanisms involve various angles of dihedrals, as shown in Fig. 1. CP, which transmits circularly polarized signals, is less capable of distinguishing between dihedrals of different angles, resulting in less information being obtained compared to DP modes. DP, which transmits linearly polarized signals, is sensitive to urban areas containing dihedrals at different angles. Particularly in urban environments where buildings have various orientations but the angles within each building cluster are consistent, as shown in Fig. 2, DP has a significant advantage in distinguishing buildings with different orientations.

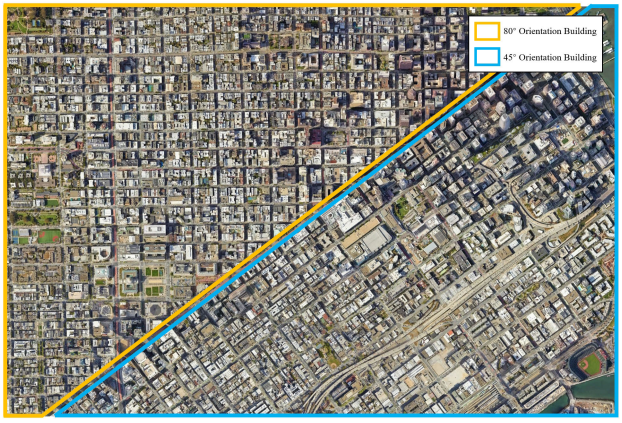


Fig. 1. Optical images of buildings at different angles derived from Google Earth in San Francisco area.

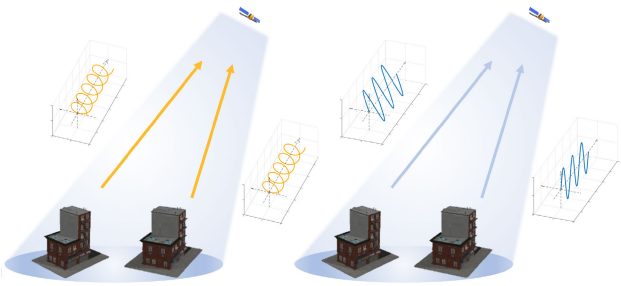


Fig. 2. Schematic when SAR transmits circular polarization and linear polarization separately, circular polarization in the received signal is difficult to distinguish dihedrals at different angles, while linear polarization can distinguish them.

In vegetated areas, the situation is more complex compared to urban regions. For forests and other vegetated areas dominated by volume scattering, the scatterers are typically a mix of dihedrals at different angles and dipoles. Neumann et al. [51] modeled forests using spherical and ellipsoidal scatterers with random orientations. In practice, the specific types and growth conditions of vegetation need to be analyzed case by case. For typical forests or woodlands, the main scattering mechanisms include: (II) low-order multiple scattering, (III) random volume backscatter from the impenetrable layer of discrete scatterers, and (V) single scattering from anisotropic structures. For vegetation lacking dense canopies, canopy-related scattering (III) is diminished; for crops without tall trunks, low-order multiple scattering (II) is also relatively weaker. Therefore, the scattering characteristics of specific vegetation types require careful consideration.

In the classification of vegetated areas like forests, DP modes are more sensitive to the orientation of dihedral scatterers, resulting in greater inconsistency in echo characteristics from different locations. CP modes cannot distinguish rotating dihedrals and helices, resulting in relatively consistent

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echo characteristics and demonstrating the rotational invariance of CP-SAR. This consistency in echo characteristics is crucial for the Wishart classifier, which models the covariance matrix assuming a Wishart distribution and relies on the physical scattering similarity between classes for classification. More consistent echo characteristics lead to smaller within-class distances. Therefore, in the classification of vegetated areas like forests, CP often provides better classification performance due to its consistent echo characteristics.

Overall, when dealing with complex urban areas, CP struggles to differentiate buildings at various angles. In certain vegetated areas, CP shows smaller Wishart within-class distances. This difference in the ability of different polarimetric SAR modes to distinguish between helices and angled dihedrals leads to CP modes showing a clear advantage in the classification of azimuthally symmetric natural targets. Conversely, in environments exhibiting azimuthal asymmetry, such as urban areas, DP modes demonstrate their superiority.

III. Optimal Classification Polarization Mode Selection Parameters

Based on the conclusion in Section II, it can also be inferred that there are only differences in the transmit polarization types among the various GCP modes. The most effective GCP mode for different classification applications may vary. Sometimes, the optimal mode may be found in elliptical polarization modes other than DP or CP modes. This demonstrates the applicability and flexibility of GCP modes.

Considering that the optimal polarization mode for different classification application scenarios may be elliptical polarization, to identify the most effective and optimal GCP polarization mode for classification applications, this article proposes optimal classification polarization mode selection parameters based on the generalized WMM (GWMM) classification method.

A. GWMM Classification Method

Raney [52] pointed out that studies using inappropriate polarization parameters or pseudo-covariance matrix expansions should be ignored, as these comparisons fundamentally arise from a flawed methodology. Most critics of CP defend their stance by citing poor results obtained through such methods.

We use the GWMM method proposed by Hou et al. [44] for classification. This method extends the WMM to avoid the issue of lacking a unified standard for feature parameter extraction between different modes in other classification methods. The method skips the feature extraction process and directly bases classification on the statistical characteristics of the coherence/covariance matrix, which contains all the polarization information obtainable by the respective system. This allows QP, CP, and DP modes to use all the polarization information they have obtained within a unified framework for classification. We extend GWMM to the GCP modes.

In polarimetric SAR image processing, the Wishart distribution is often used to model multi-look polarimetric data. The Wishart distribution is a probability distribution used to describe covariance matrices, applicable to cases of multivariate Gaussian distribution, and is suitable for typical polarimetric SAR images. The Wishart distribution can be used to describe the statistical characteristics of the multilook polarimetric scattering matrix of each pixel in SAR images. Each pixel's scattering matrix can be considered a superposition of contributions from multiple individual scattering centers, each of which can be described by a Wishart distribution. Therefore, a multilook polarimetric scattering matrix can be considered a mixture of multiple Wishart distributions. The probability density function of the Wishart distribution for the L-look covariance matrix T of polarimetric SAR images is given by

$$\mathbf{p}(\mathbf{T}|\mathbf{Z}) = \frac{L^{Lq}|T|^{L-q}}{\Gamma_q(L)|Z|^L}\exp(-L\cdot\mathrm{tr}(Z^{-1}T)) \tag{11}$$

where Z is the distribution center, q is the dimension of T, and $\Gamma_q(L)$ is a constant, which can be calculated using the following equation:

$$\Gamma_{\mathbf{q}}(\mathbf{L}) = \pi^{\mathbf{q(q-1)}/2}\prod_{j=1}^{q}\Gamma(L-j+1). \tag{12}$$

In the above equation, Γ is the standard gamma function.

Wishart distance refers to the distance between two random matrices under the Wishart distribution. In polarimetric SAR image classification, each pixel's polarimetric covariance matrix is considered a random matrix. By calculating the Wishart distances between these random matrices, different classes of pixels can be distinguished. First, the average covariance matrix of known-category training samples is calculated, yielding the mean and covariance of Wishart distances for each category. Next, the Wishart distance of the pixel to be classified is compared with the mean distance of each category, and the pixel is assigned to the category with the smallest Wishart distance. Lee et al. [53] derived the Wishart distance from the log-likelihood of the Wishart distribution. The corresponding form under the covariance matrix is as follows:

$$d_W(T_m, T_n) = \ln|T_n| + \mathrm{tr}\left(T_n^{-1}T_m\right) \tag{13}$$

where T_m and T_n represent the covariance matrix of any two samples.

The core of Hou's classification method is to equivalently replace the covariance matrix in Wishart classification with Stokes vectors that all GCP polarization patterns have

$$R(\mathbf{G}_m, \mathbf{G}_n) = \frac{1}{4}(g_{m0}g_{n0} - g_{m1}g_{n1} - g_{m2}g_{n2} - g_{m3}g_{n3}). \tag{14}$$

So as to calculate the Wishart distance

$$d_w(\mathbf{G}_m, \mathbf{G_n}) = \ln[R(\mathbf{G_n}, \mathbf{G_n})] + 2\frac{R(\mathbf{G_m}, \mathbf{G_n})}{R(\mathbf{G_n}, \mathbf{G_n})}. \tag{15}$$

In the GWMM classification method, distances are mapped to similarity functions, and each category is subdivided into several subclasses to more precisely model the distribution characteristics of real SAR images, thereby improving classification performance. A detailed description of this method is beyond the scope of this article and will not be elaborated here.

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B. Total Relative Wishart Distance Difference and Average Relative Wishart Distance Difference Within the Class

To identify the optimal polarization mode with the highest classification accuracy within the GCP modes, this article proposes two parameters to evaluate which mode has better classification advantages.

1) *Parameter for the Entire Scene: Total Relative Wishart Distance Difference N_d*

$$N_d = \sum_{i=1}^{N} \left(\frac{Q_{1,i} - Q_{2,i}}{Q_{1,i}} \right). \tag{16}$$

In the above equation, Q is the comprehensive Wishart distance obtained by weighting the pixel with n subclasses and mapped to the similarity parameter defined by the kernel function. $Q_{1,i}$ and $Q_{2,i}$ represent the maximum similarity probability and second maximum similarity probability corresponding to the nearest and second nearest Wishart distances, respectively, representing the two most likely categories that the pixel belongs to. Divide by the nearest Wishart distance $Q_{1,i}$ to become a relative value to eliminate the influence of the absolute size of the Wishart distance itself for different polarization modes. Fundamentally, $Q_{1,i}$ and $Q_{2,i}$ correspond to the smallest and second-smallest Wishart distances computed in the GWMM classification method. These Wishart distances are mapped to similarity parameters Q using a Laplacian kernel function. The computation is as follows:

$$Q = \sum_{i=1}^{N} e^{(-\beta/d_{wi})} \cdot \pi_i \tag{17}$$

where N is the number of subclasses, β is a positive hyperparameter, d_{wi} is the Wishart distance for subclass i, and π_i is the weighting coefficient of subclass i.
The significance of this parameter is that the relative value of this probability reflects the classification performance of the pixel in the current polarization mode. The larger the parameter, the greater the difference between the pixel and the second similar category corresponding to $Q_{2,i}$, making it easier to classify to the accurate category corresponding to $Q_{1,i}$. The greater the discrimination, the better the classification effect. The larger the value of N_d, the greater the overall discrimination, which means that the classification effect of the entire scene is better in this polarization mode.

2) *Parameter for Specific Class Classification Tasks: Average Relative Wishart Distance Difference Within the Class $\bar{N}_c$*

$$\bar{N}_c = \frac{1}{M} \sum_{j=1}^{M} \left(\frac{Q_{1,j} - Q_{2,j}}{Q_{1,j}} \right). \tag{18}$$

In the above equation, $Q_{1,j}$ and $Q_{2,j}$ represent the maximum and second maximum similarity probabilities corresponding to the nearest and second-nearest Wishart distances for a pixel in the classification task, respectively. These probabilities reflect the two most likely categories to which the pixel belongs. Dividing by $Q_{1,j}$ normalizes the values, and dividing by M, the number of correctly classified pixels, averages the values. The significance of this parameter is as follows: the difference between these two probabilities, divided by the maximum probability, reflects the classification performance of the pixel under the current polarization mode. Averaging over the number of pixels in the same category gives the separability of the pixel between the two leading classes in the classification. The larger the separability, the less likely it is that a category will be misclassified as the closest second category during classification, and the more accurate the classification. The larger the value of $\bar{N}_c$, the better the performance of the current polarization mode for the specific classification task.

These two parameters can effectively measure the classification performance of various polarization modes in the target classification task and predict which GCP mode is optimal. The specific experimental results will be demonstrated in Sections IV-C2 and IV-C3.

IV. Results

In this section, we use satellite images of the San Francisco area to validate the above analysis and proposed parameters. In Section IV-A, the datasets are introduced. In Section IV-B, we compare the classification performance of different polarization modes. Section IV-C discusses the effectiveness of the proposed parameters in evaluating the classification performance of different polarization modes.

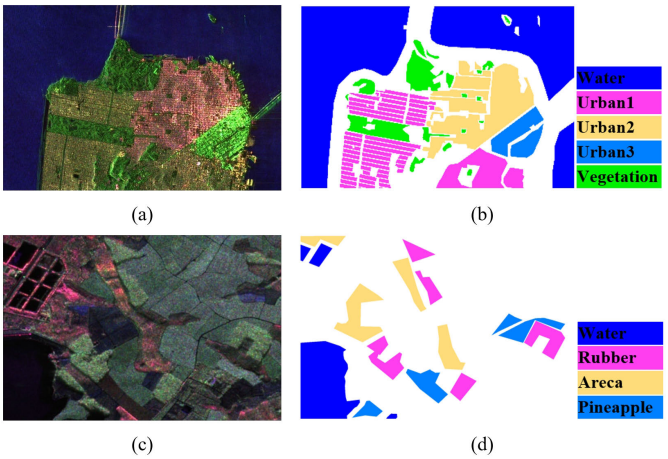
A. Datasets

The first dataset is the GF-3 C-band QP satellite image of the San Francisco area. Fig. 3(a) and (b) shows the Pauli color-coded image and the ground-truth image, respectively. The white areas in Fig. 3(b) indicate regions where ground truth is unavailable. The color legend is displayed on the right side of the ground-truth image. The image size is 1600 × 2500 pixels, and the dataset consists of five categories: water, urban1 with a 90° orientation angle, urban2 with an 80° orientation angle, and urban3 with a 45° orientation angle, and vegetation.

The second dataset is L-band airborne QP SAR data obtained by the Aerospace Information Research Institute, Chinese Academy of Sciences, in Hainan Province in 2021. The image size is 1500 × 1000 pixels, and the dataset consists of four categories: water, rubber trees, areca trees, and pineapples. Fig. 3(c) and (d) shows the Pauli color-coded image and the ground-truth image, respectively.

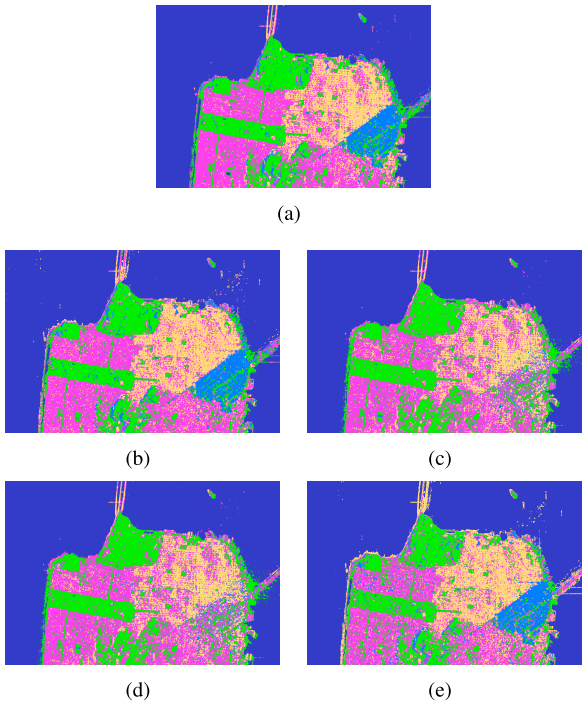
After obtaining the 5 × 5 multi-look covariance matrix of QP data, in the absence of real GCP data, QP data can be used to simulate the Stokes vector of GCP based on first principles [45] and GCP data can be simulated using (5). During the training process, only 1% of the true pixels are used as training samples, and the rest are used as test samples.

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(a) (b) (c) (d)

Fig. 3. Dataset for the experiment. (a) Pauli color images from the San Francisco area dataset. (b) Ground-truth images from the San Francisco area dataset (five categories). (c) Pauli color images from the Hainan Province dataset. (d) Ground-truth images of the Hainan Province dataset (four categories).



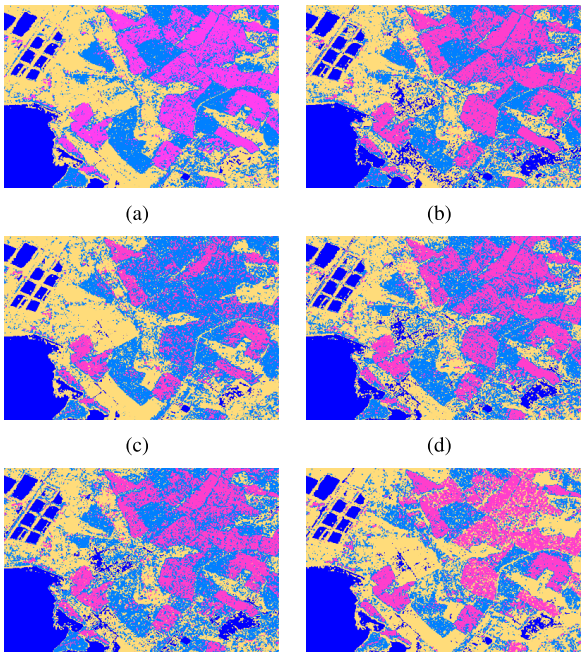
(a) (b) (c) (d) (e)

Fig. 4. Classification images of five different polarimetric modes on the San Francisco dataset. The polarization modes in (a)–(e) are QP, H, L, R, V, respectively. Here, H represents the DP mode transmitting H polarization (HH, VH), R represents the CP mode transmitting R polarization (HR, VR), and V and L are defined similarly. The same notation is used throughout the following text.

B. Classification Performance

To validate the aforementioned points, the GWMM method was used to classify 399 types of GCP modes, including traditional CP and DP.

1) CP and DP: Considering that the San Francisco area includes the three most typical feature types in natural areas, we first performed GWMM classification for traditional CP and DP modes in the San Francisco area. The classification results are shown in Figs. 4 and 5.



(a) (b) (c) (d) (e) (f)

Fig. 5. Classification images of six different polarimetric modes on the Hainan dataset. The polarization modes in (a)–(f) are QP [40, −5] H L R V, respectively. [40, −5] Means 40° elliptical orientation angle and −5° ellipticity angle polarization mode. The same notation is used throughout the following text.

TABLE III
CLASSIFICATION ACCURACY OF DIFFERENT POLARIZATION MODES IN SAN FRANCISCO DATASET

Classes	QP	H	V	R	L
Water	99.97	99.98	99.99	**99.99**	99.99
Urban1	72.35	**72.35**	72.27	62.99	62.57
Urban2	86.68	**84.97**	84.15	75.52	75.43
Urban3	89.72	**85.60**	72.79	35.15	36.67
Vegetation	79.87	**79.87**	67.71	75.29	74.75
OA	90.80	**90.49**	89.05	85.62	85.66

We selected urban areas with different orientation angles in the San Francisco dataset. The classification results Fig. 6 show that for the three urban areas with different angles (urban1, urban2, and urban3), the pink-hued DP mode achieves higher classification accuracy compared to the purple-hued CP mode. The overall classification accuracy (OA) is also higher for DP.

Combining the analysis of the typical target scattering matrices in Section II-C, it is difficult to distinguish dihedral scattering with certain angles when transmitting circularly polarized waves, whereas linearly polarized waves can effectively distinguish this feature. When the building walls are vertical to the ground, they form dihedral reflectors. If the radar's illumination direction is perpendicular to the walls and the ground, strong double-bounce scattering echoes are produced. Since the streets and buildings along them in urban areas typically form a grid-like pattern, these regions are usually dominated by double-bounce scattering. In building

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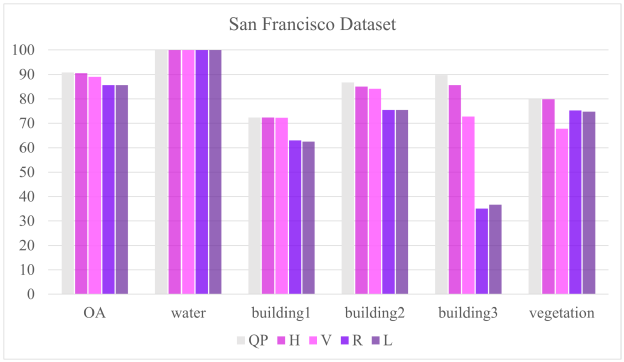


Fig. 6. Classification accuracy of different polarization modes in San Francisco dataset.

areas using GF-3 satellite data (with a resolution of 8 meters), the echo signal is formed by the coherent synthesis of scattering centers within the resolution cell. However, circular polarization performs poorly in distinguishing dihedral echoes with specific angle differences, leading to overly similar echo signals from building areas with different angles, making them difficult to distinguish. This is clearly evident from the classification results as shown in Table III: for urban1, urban2, and urban3 categories with angles of 90°, 80°, and 45°, respectively, the classification performance of circularly compact polarimetry is not as good as that of DP.

In Hainan dataset, we selected three different crops. The classification results Fig. 7 show that for the three different crops, the purple-hued CP mode achieves higher classification accuracy compared to the pink-hued DP mode. The OA is also higher for CP. We also added another GCP mode with an elliptical orientation angle of 40° and an ellipticity angle of −5°. In the following text, the GCP mode is represented in the form of examples such as [40, −5], the first number in [] represents the elliptical orientation angle, and the second number represents the ellipticity angle. It can be seen that the classification accuracy of this polarization mode in each crop area is higher than that of H or V or R or L, and the OA is also higher. Obviously, this GCP mode is more suitable for the current classification scenario. In fact, this mode is the GCP mode with the highest OA in the current scenario, which will be explained in Section IV-C1. However, although this GCP mode has the highest OA, it does not achieve the best classification performance for each class. For the three crop areas, there are still two classes where the classification accuracy of the other two GCP modes is higher than that of the GCP mode with an elliptical orientation angle of 40° and an ellipticity angle of −5°. From Table IV, it can be seen that the accuracy of the rubber category in the [45, −15] mode is higher than that in the [40, −5] mode and the accuracy of the areca category in the [30, 5] mode is higher than that in the [40, −5] mode.

2) GCP: To classify GCP modes, considering the existence of numerous GCP modes, we first classify the GCP modes that can be currently implemented in real systems. These include circular polarization, linear polarization, and elliptical polarization, with the latter's orientation angle limited to ±45° and the ellipticity angle varying from +45° to −45°. The

TABLE IV
CLASSIFICATION ACCURACY OF DIFFERENT POLARIZATION MODES IN HAINAN DATASET

Classes	QP	[40 , -5]	[45 , -15]	[30 , 5]	H	V	R
Water	100.00	99.96	99.99	99.91	**100.00**	100.00	100.00
Rubber	90.76	86.07	**86.86**	85.44	84.77	81.74	83.78
Areca	97.81	93.73	93.12	**94.92**	93.48	85.60	90.57
Pineapple	78.29	**70.43**	67.89	68.93	58.81	53.42	66.44
OA	93.63	**90.21**	89.64	89.90	86.77	83.13	88.18

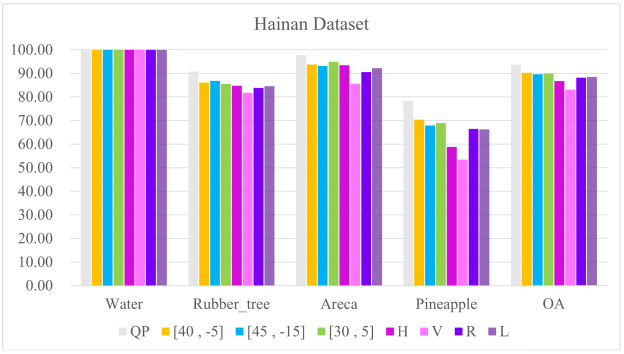


Fig. 7. Classification accuracy of different polarization modes in Hainan dataset.

classification results are shown in Fig. 8, along with the reference results for the QP mode.

From the classification results, it is evident that different polarization modes exhibit varying performance for specific categories.

1) *Water Category:* HV linear polarization and RL circular polarization demonstrated the best classification performance, both achieving classification accuracies above 99%. The $\pm\pi/4$ mode performed the worst. The elliptical polarization, fixed at ±45° orientation, improved in classification performance with increasing ellipticity.
2) *Building1 Category:* HV performed the best, comparable to the QP mode. −45° linear polarization performed slightly better than 45° linear polarization. RL circular polarization had the worst performance among the 22 modes. Among elliptical polarizations, the 40° ellipticity angle mode performed as poorly as the RL circular polarization mode. The 10° ellipticity angle mode, which resembles linear polarization, performed better than the 20°, 30°, and 40° modes, with classification accuracy decreasing in that order.
3) *Building2 Category:* HV performed the best, with classification accuracy almost 10% higher than the second-best RL circular polarization. The $\pm\pi/4$ mode's classification accuracy was inferior to RL circular polarization. Elliptical polarization accuracy transitioned from resembling the $\pm\pi/4$ mode to resembling circular polarization, showing an increase from low to high. Ellipticity angles of 30° and 40° performed similarly to RL circular polarization.
4) *Building3 Category:* H polarization had the highest classification accuracy, exceeding the second-best V polarization by 13%. RL circular polarization was

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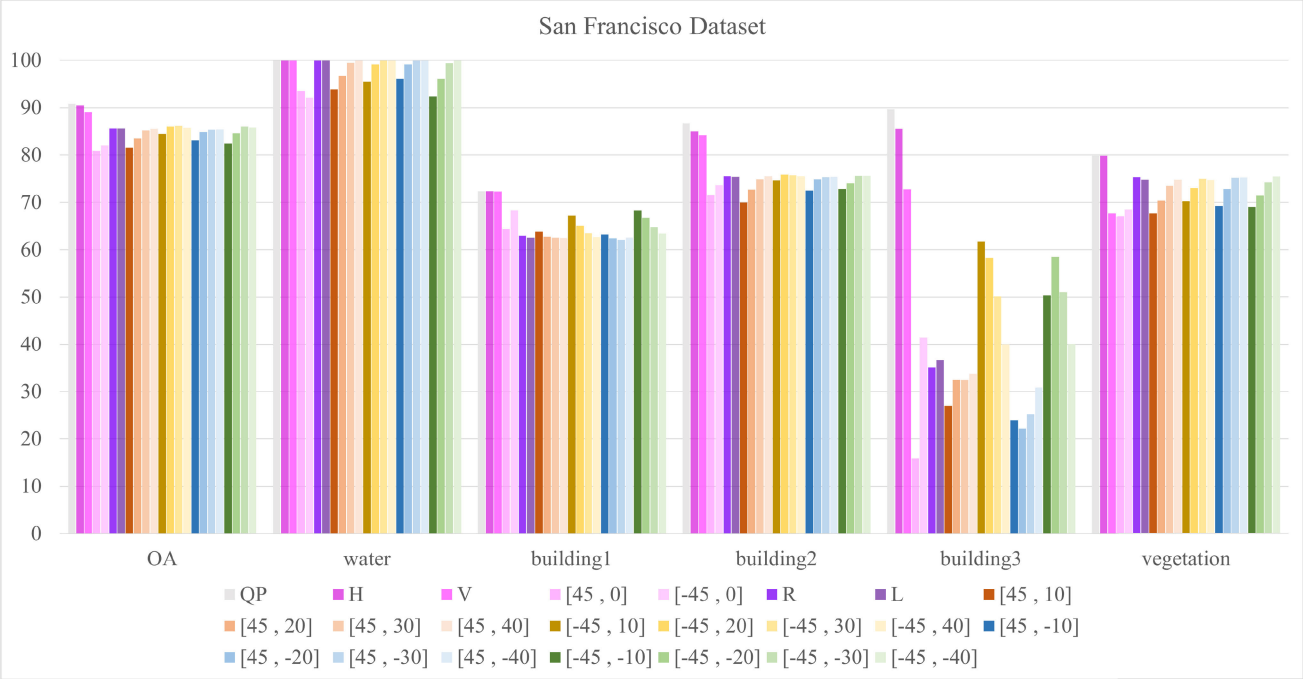


Fig. 8. Classification results of GCP modes achievable by current systems. Each color system represents GCP modes with the same orientation angle and different ellipticity angles. The lighter the color, the more circular the polarization mode.

inferior to $-45°$ linear polarization but better than $45°$ linear polarization. Elliptical polarizations with $-45°$ orientation angles had higher classification accuracy than RL circular polarization. The best-performing elliptical polarization with a $10°$ ellipticity angle outperformed the L mode by 25% but still trailed the best H polarization by 24%.

5) *Vegetation Category:* H polarization had the highest accuracy, closely followed by the RL mode. The V and modes performed the worst. Some elliptical polarization modes close to circular polarization had classification accuracies similar to RL.

These results indicate that selecting the most suitable polarization mode for specific classification tasks is crucial. Elliptical polarization sometimes outperforms circular compact polarization in specific tasks, such as classifying buildings with angles (e.g., Building3 category with primarily volumetric scattering mechanisms). This phenomenon is reasonable because, with varying ellipticity angles, elliptical polarization modes gradually transition from circular to linear polarization, reducing the difficulty in distinguishing dihedral angles caused by circular polarization.

In Section IV-C, we will present the results of our proposed two optimal classification polarization mode selection parameters: total relative Wishart distance difference and average relative Wishart distance difference within the class. These parameters will help measure the classification accuracy differences induced by different polarization modes and identify the best-performing polarization mode.

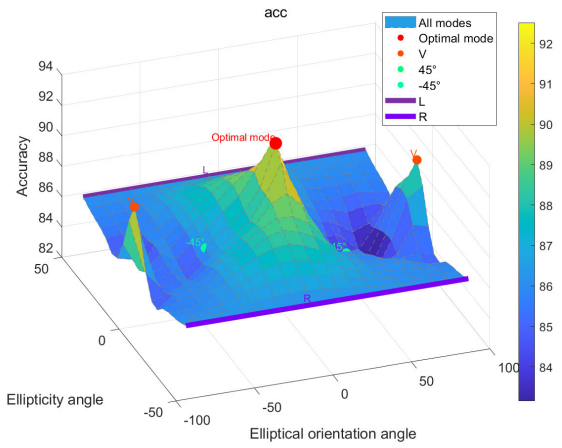
C. Classification Performance Evaluation Parameter Results

To comprehensively identify the optimal GCP mode, we simulated GCP scenarios with elliptical orientation angles ranging from $-90°$ to $90°$ in $10°$ increments, and ellipticity angles ranging from $-45°$ to $45°$ in $5°$ increments, along with $\pm 45°$ linear polarization. This resulted in a total of 399 polarization modes to observe the classification performance of GCP modes and evaluate the scene classification performance parameters.

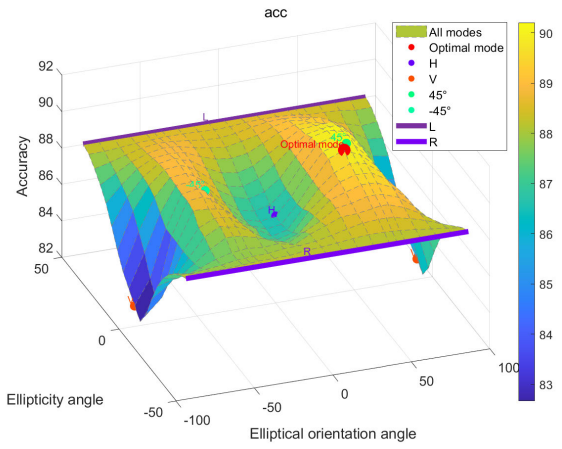
1) Classification Results of 399 GCP Modes: As can be observed from the results Fig. 9, the OA 3-D plot effectively presents the classification accuracy trend of each polarization mode, making it easy to visualize how the OA changes with the elliptical orientation and ellipticity angles. Fig. 9(a) mainly shows the results for the San Francisco area, which is dominated by buildings: the optimal classification mode for this scenario is H-polarization, HV-polarization forms a local peak within its range, and the classification accuracy of polarization modes similar to HV is generally high. RL polarization at the front and rear edge positions does not have a particularly significant advantage in classification accuracy, but it is better than some elliptical polarization modes and $\pm\pi/4$ modes.

Fig. 9(b) shows the results for the Hainan area, which is mainly dominated by crops: the optimal polarization mode is the GCP mode with an ellipticity angle of $40°$ orientation angle and $-5°$ ellipticity angle. HV polarization forms a concave area with the worst classification accuracy, while RL polarization shows moderate performance. The area around the optimal GCP mode has the best classification accuracy, with other GCP modes exhibiting varying levels of performance. This result is also consistent with our analysis in Section II, which shows that the HV polarization mode that can distinguish between different dihedral angles has the best classification effect in the San Francisco area, which is mainly dominated by buildings. This is shown in the classification accuracy plots Fig. 10 for each category in the San Francisco area below [building areas

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(a)



(b)

Fig. 9. (a) and (b) 3-D plots of OA for GCP modes in San Francisco and Hainan areas, respectively. The x- and y-axes represent the elliptical orientation angle and ellipticity angle, respectively, and the z-axis represents the classification accuracy. As shown in the legend in the upper right corner, the optimal mode and the classic HVRL and 45° polarization modes are marked on the plot.

Fig. 10(b)–(d)], while in the Hainan area, which we have investigated in the field and is mainly dominated by vegetation, a certain GCP mode with the best response is dominant.

Figs. 10(a) and 11(a) show that the classification results for the water body category are very consistent for both datasets. Both HV and RL polarization have good performance, while mode has the worst performance. The remaining GCP modes also have their own advantages and disadvantages. Our results extend the conclusion of Mittal and Singh, who found that circular polarization is the best choice for distinguishing water bodies [54].

The detailed classification accuracy results for different categories in Hainan reveal that even within the same scene, the optimal mode can vary between categories. For example, the best mode for rubber trees [second category shown in Fig. 11(b)] is the GCP mode with a 45° orientation angle and −15° ellipticity angle, while the best mode for areca trees [third category shown in Fig. 11(c)] is the GCP mode with a 30° orientation angle and 5° ellipticity angle. For pineapples [fourth category shown in Fig. 11(d)], the optimal mode is the GCP mode with a 40° orientation angle and −5° ellipticity angle. These differences can be attributed to the inherent characteristics of the crop areas. For instance, the poor performance of HV polarization in the pineapple region might be due to the inclined orientation of pineapples, which affects the performance of modes like $\pm\pi/4$ that may otherwise perform poorly in certain scenes but excel in others, such as the pineapple area. This can be clearly seen in the classification accuracy plots, where the best performance is achieved near the $\pm\pi/4$ mode in the optimal GCP mode.

2) Total Relative Wishart Distance Difference: The proposed parameter, the total relative Wishart distance difference, is used to identify the optimal GCP polarization mode for overall classification performance. This parameter calculates the sum of the Wishart distance differences between the closest and second-closest categories for every pixel in the entire image. For an individual pixel, a larger Wishart distance difference between the closest and second-closest categories indicates a higher distinction between the pixel and its confusing categories under the current polarization mode. By performing this calculation for the entire image and adjusting for inconsistencies in the absolute values of Wishart distances across different polarization modes, the total relative Wishart distance difference provides a robust metric for evaluating the separability of the entire scene. Therefore, a larger sum indicates a higher overall separability for the polarization mode, leading to better classification performance. The parameter results are shown in Fig. 12.

Fig. 12(a) and (b) displays the results for San Francisco and Hainan, respectively, arranged in descending order of classification accuracy. The parameter results closely match the classification accuracy, effectively measuring the classification performance of different polarization modes. The polarization mode with the highest total relative Wishart distance difference corresponds to the highest classification accuracy, consistent with our analysis. This confirms that the parameter is indeed capable of identifying the polarization mode with the highest OA for the current scene.

In the top 20 polarization modes for classification accuracy, the parameter results perfectly align with the classification performance for both scenes. However, minor fluctuations occur in the later modes, likely due to the increased number of misclassified pixels in these less effective modes. As the number of misclassified pixels grows, the difference between the closest and second closest Wishart distances for these pixels increases, not reflecting higher classification accuracy but rather indicating greater divergence in misclassification. This leads to larger parameter values and some fluctuations. Nonetheless, the impact of these fluctuations is minimal and does not affect the ability of the total relative Wishart distance difference to identify the polarization mode with the optimal classification performance and highest overall separability.

3) Average Relative Wishart Distance Difference Within the Class: The proposed parameter, the average relative Wishart distance difference within the class, is used to identify the optimal polarization mode for classifying a specific category. This parameter measures the separability of a specific category's

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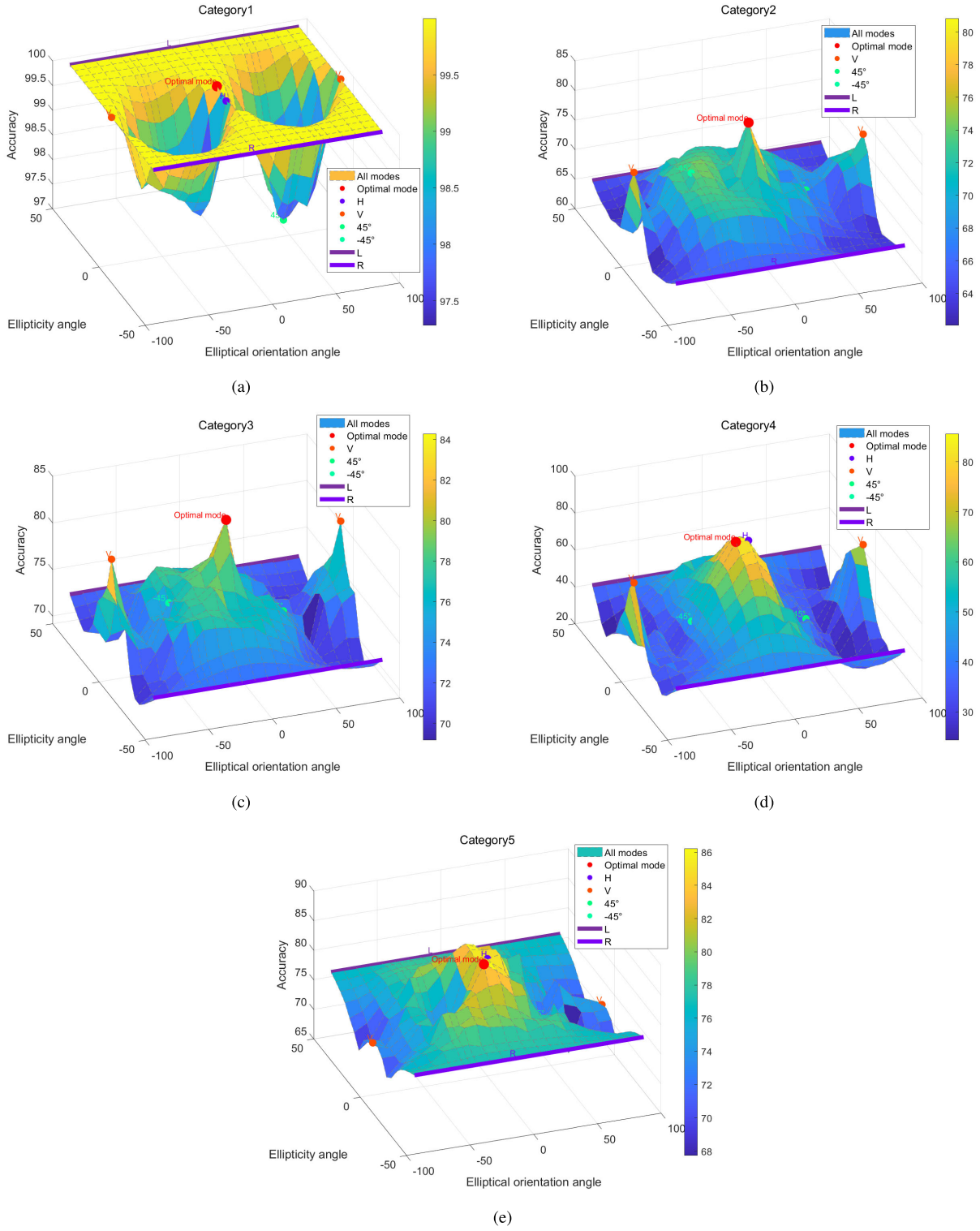


Fig. 10. (a)–(e) Classification accuracy of 399 GCP modes for five categories in the San Francisco area: water bodies, buildings with orientation angles of 90°, 80°, and 45°, and vegetation.

pixels from pixels of other categories under a given polarization mode. By averaging the sum of relative Wishart distance differences, it calculates the average separability of each pixel within the category, providing a metric for evaluating the classification performance of each pixel under the current polarization mode. The significance of this parameter is clear: the larger the value, the higher the separability of the category from other categories under the current polarization mode, thus indicating better classification performance.

The results in Figs. 13 and 14 show that the proposed parameter, the average relative Wishart distance difference within the class, closely matches the classification results and

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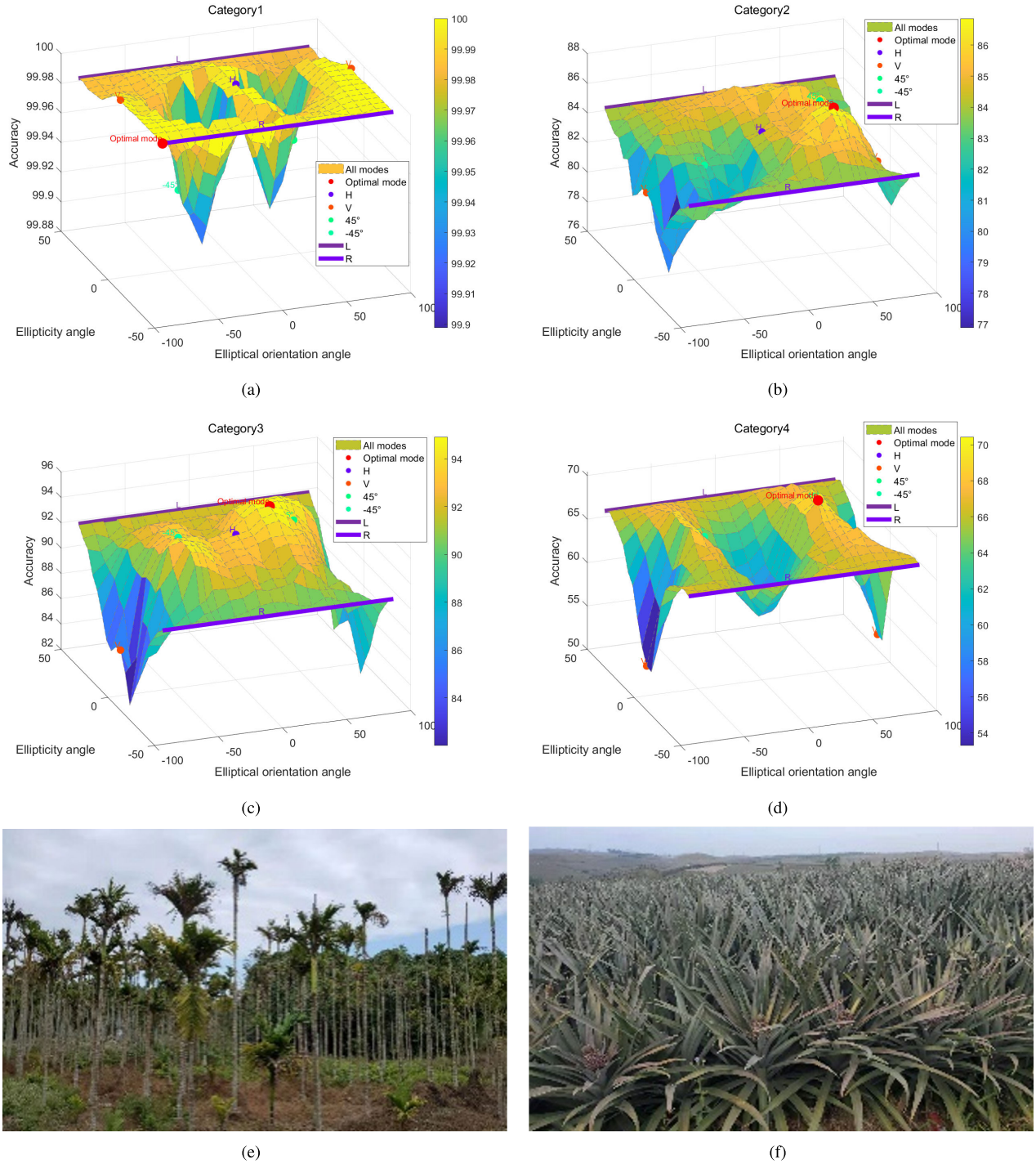
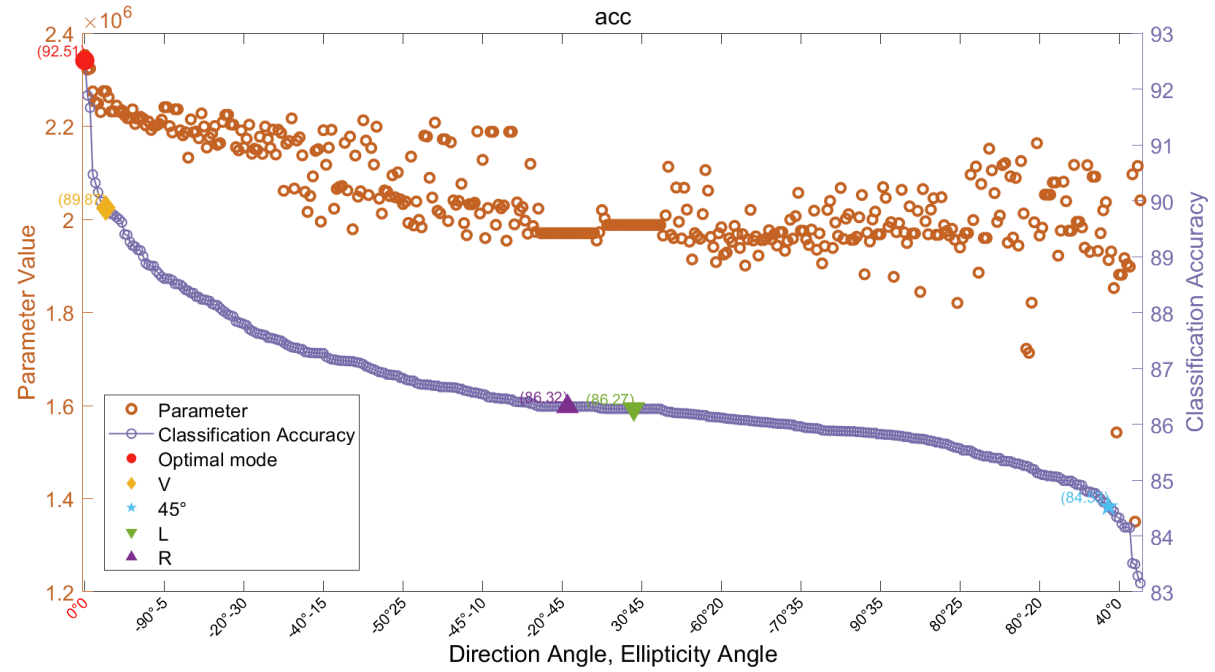


Fig. 11. (a)–(d) Classification accuracy of 399 GCP modes for four categories in the Hainan area: water bodies, rubber trees, areca trees, and pineapples, (e) and (f) images of areca trees and pineapple fields that we surveyed in the field.

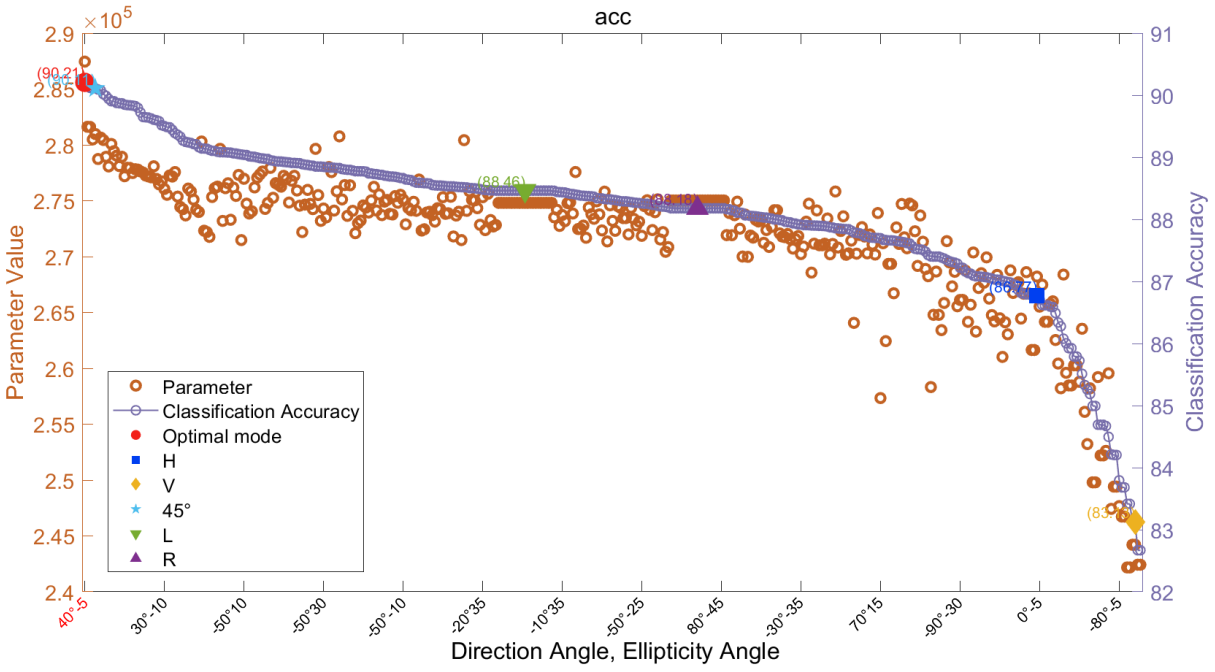
effectively reflects the changes in classification performance of different GCP modes.

1) Figs. 13(a) and 14(a) display the results for the first category, water bodies. The performance of the GCP modes with classification accuracies over 99.9% is very stable for the first 200 modes. The parameter values for less effective modes are relatively low. The fluctuations in Fig. 14(a) are more pronounced due to the small differences between the modes, with all classification accuracies being over 99%.
2) Fig. 13(b) presents the results for the second category, Building1, and Fig. 13(c) for the third category, Building2. The parameter results for these two categories are closely related to the classification performance of each polarization mode. For the top ten polarization modes with the best classification performance, the proposed parameter shows perfect consistency. The maximum parameter value corresponds to the optimal polarization mode, which is the H polarization (0° orientation angle and 0° ellipticity angle).

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(a)



(b)

Fig. 12. (a) and (b) Classification accuracy and total relative Wishart distance difference parameter values for San Francisco and Hainan areas, respectively. The values are sorted in descending order of classification accuracy. Each mode corresponds to a classification accuracy and a parameter value. The x-axis represents the polarization mode, the orange left axis represents the proposed parameter value, and the purple right axis represents the classification accuracy. Refer to the legend in the upper right corner for the positions of the classification accuracy of some typical polarization modes among the 399 GCP modes.

3) Fig. 13(d) shows the results for the fourth category, Building3. The proposed parameter almost perfectly corresponds to the top 50 GCP modes with the best classification performance. The maximum parameter value corresponds to the best-performing GCP mode ($-10°$ orientation angle and $0°$ ellipticity angle). However, after the classification accuracy drops from the best 85% to 60%, the proposed parameter exhibits significant fluctuations, no longer corresponding well with the classification performance. This is mainly because Building3 represents strongly volume scattering inclined buildings, where certain specific pixels with particularly strong

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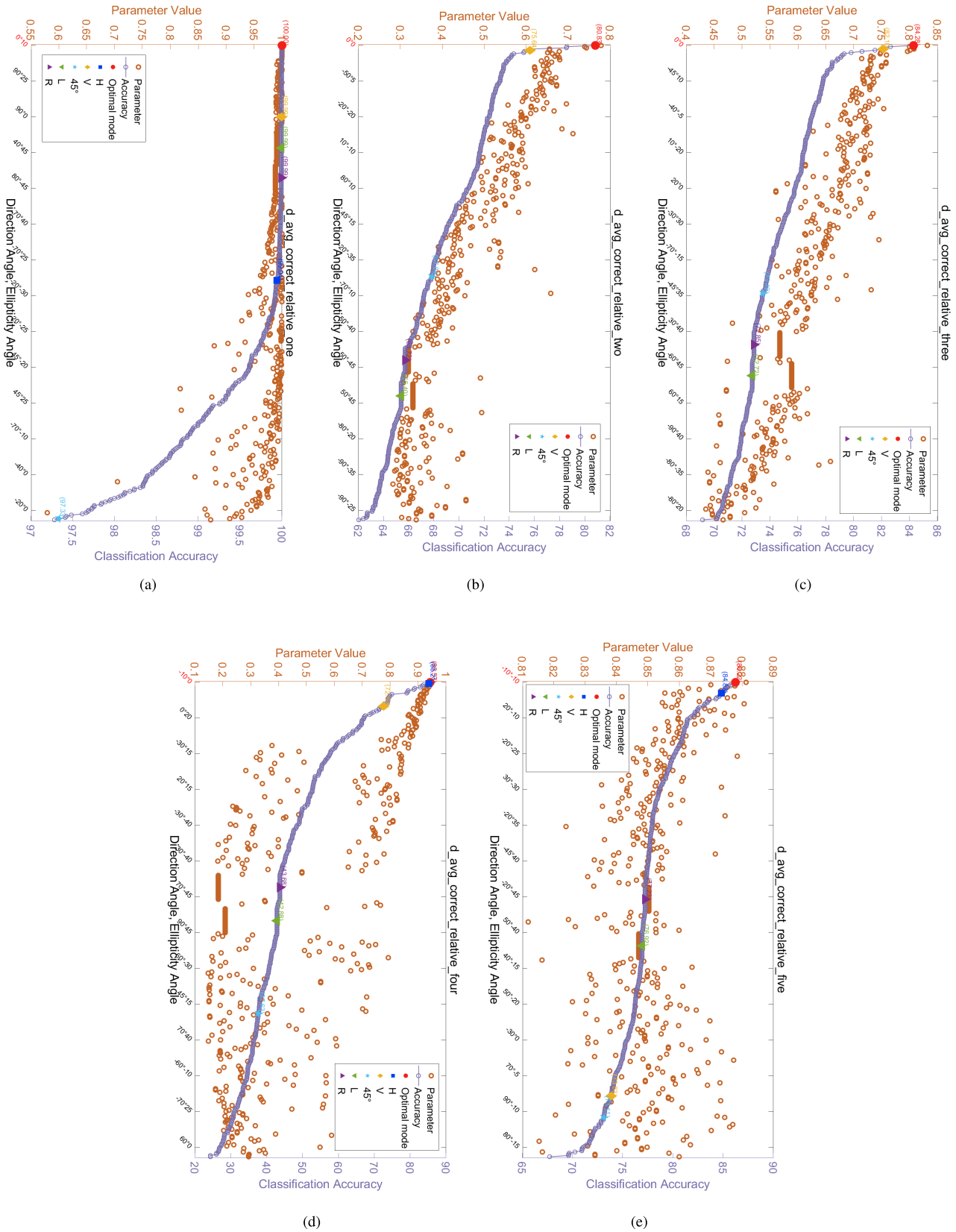


Fig. 13. (a)–(e) Results for the categories 1–5 in San Francisco area, respectively. The values are sorted in descending order of classification accuracy. Each mode corresponds to a classification accuracy and a parameter value. The x-axis represents the polarization mode, the orange left axis represents the proposed parameter value, and the purple right axis represents the classification accuracy. Refer to the legend for the positions of the classification accuracy of some typical polarization modes among the 399 GCP modes.

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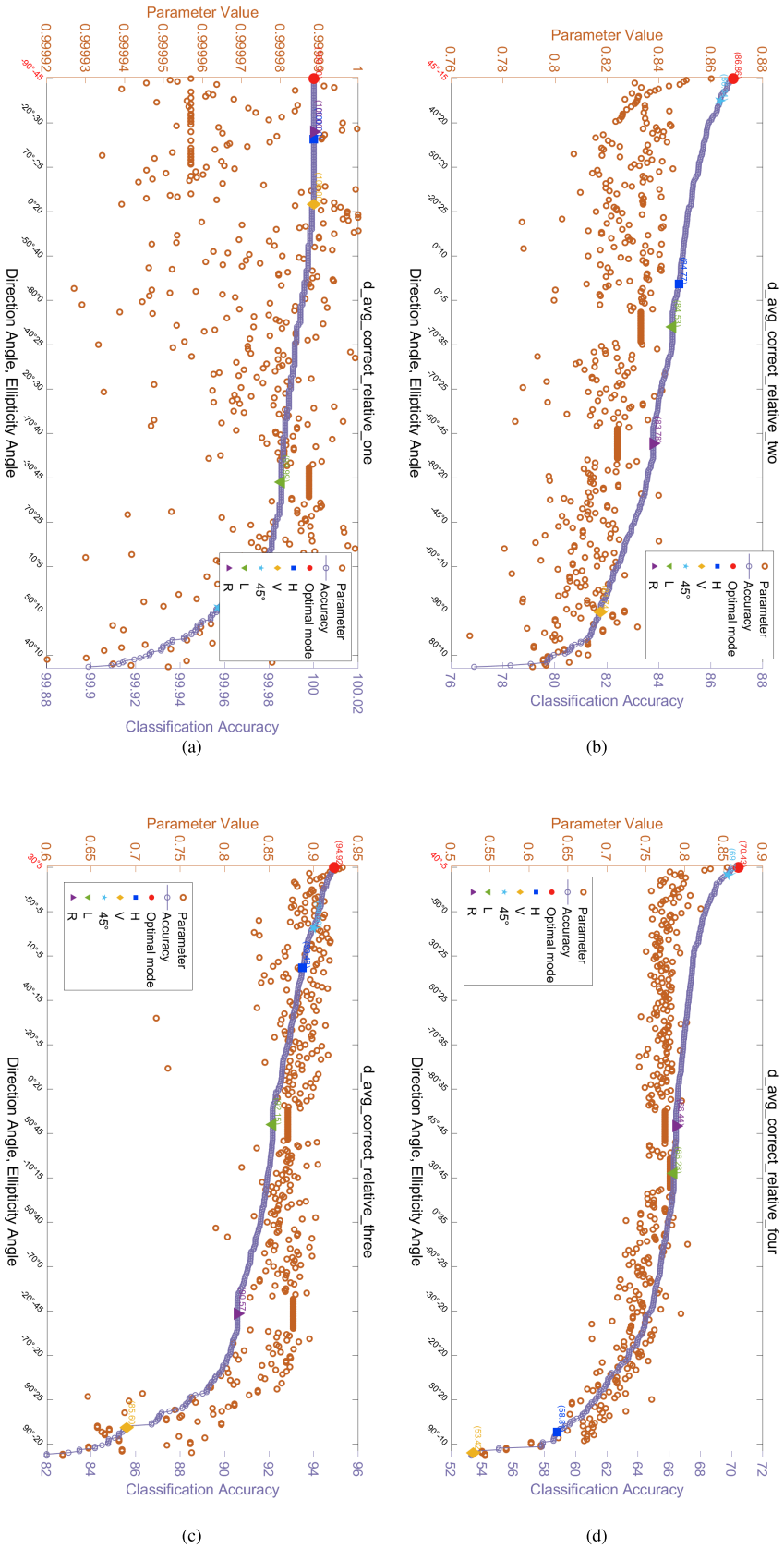


Fig. 14. (a)–(d) Results for the categories 1–4 in Hainan area, respectively. The values are sorted in descending order of classification accuracy. Each mode corresponds to a classification accuracy and a parameter value. The x-axis represents the polarization mode, the orange left axis represents the proposed parameter value, and the purple right axis represents the classification accuracy. Refer to the legend for the positions of the classification accuracy of some typical polarization modes among the 399 GCP modes.

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scattering have a significant weight in the parameter. These few pixels cannot represent the classification performance of the entire category, and this issue is more pronounced when the OA is low, causing the parameter to misalign with the classification accuracy. Despite the fluctuations, the overall trend still shows consistency between the proposed parameter and the classification performance.

4) Fig. 13(e) shows the results for the fifth category, vegetation. This category also exhibits significant fluctuations for similar reasons to Building3, involving complex volume scattering environments including trees, grasslands, and parks. In such heterogeneous conditions, the response of different polarization modes to the parameter varies greatly. For the top 200 polarization modes in classification accuracy, only four modes show significant deviations. After excluding these four outliers, the other modes align very well with the classification accuracy, accurately identifying the optimal GCP mode as the one with a $-10°$ orientation angle and a $-10°$ ellipticity angle.
5) Fig. 14(b)–(d) displays the results for the second category (rubber trees), the third category (areca trees), and the fourth category (pineapple) in the Hainan dataset. These agricultural categories show that the proposed parameter aligns well with the classification accuracy curves and effectively measures the classification performance of GCP modes. Especially for the top 10 or 30 GCP modes with the highest classification accuracies, the parameter demonstrates near-perfect alignment and accurately identifies the optimal GCP modes for classification.

Overall, these findings indicate that the proposed average relative Wishart distance difference within the class parameter effectively identifies the optimal polarization mode for specific categories, reflecting the classification performance of different GCP modes.

V. Discussion

In this article, we first conducted a GWMM classification using quad polarization, dual-polarization, and circular compact polarization for the San Francisco area, which includes the three most typical scattering mechanisms found in nature. The classification results validated our analysis: for the water category, all modes demonstrated excellent classification performance; in urban areas, dual polarization's classification performance was nearly identical to quad polarization, while circular compact polarization performed poorly in distinguishing targets, consistent with the conclusion of Yin et al. [43] Subsequently, we classified GCP that current SAR systems can achieve. The results showed that some elliptical polarizations could achieve the same classification performance as circular compact polarization, and even better performance in Building1 and Building3 categories.

We also proposed the total relative Wishart distance difference parameter and the average relative Wishart distance difference within the class parameter to select the optimal GCP mode for classification. Experiments on 399 GCP modes for both the San Francisco and Hainan datasets indicated that the proposed parameters matched well with classification accuracy, effectively measuring polarization mode performance, and successfully identifying the optimal GCP modes. Notably, the optimal polarization modes varied for different categories in this study. For the San Francisco dataset, the optimal modes for the first three categories were H polarization, the fourth category was $-10°$ elliptical orientation angle, $0°$ ellipticity angle, and the fifth category was $-10°$ orientation angle, $-10°$ ellipticity angle, with the overall optimal mode being H polarization. For the Hainan dataset, the first category's accuracy exceeded 99%, and the optimal modes for the second, third, and fourth categories were GCP modes with orientation angles of $45°$, $30°$, and $40°$ and ellipticity angles of $-15°$, $5°$, and $-5°$, respectively, with the overall optimal mode being $40°$ orientation and $-5°$ ellipticity angle GCP mode.

It is evident that conventional HV polarization or RL polarization is not always the optimal mode. No single mode in GCP consistently performs best across all categories, aligning with our analysis. In some cases, elliptical polarization outperformed linear and circular compact polarizations, such as in the fourth category (Building3) of the San Francisco data, where the optimal mode's classification accuracy was 42% higher than the conventional circular compact polarization mode, and in the fifth category (vegetation), where the optimal mode's accuracy was 12% higher than conventional V polarization and 9% higher than circular compact polarization. For the Hainan dataset, the optimal GCP polarization modes for the three crops outperformed typical HV or RL polarization by 2% to 17%, as detailed in Table V. Therefore, using the optimal polarization mode for classification in specific scenes is crucial.

TABLE V
Comparison of Classification Accuracy Between Typical Modes and Optimal Modes

Categories	Optimal modes	H	V	R	L
Building1	H **80.8**	0	-5.2	-15.1	-15.4
Building2	H **84.3**	0	-2.2	-11.4	-11.6
Building3	[-10, 0] **85.6**	-0.4	-12.9	-41.9	-42.7
Vegetation	[-10,-10] **86.2**	-1.4	-12.4	-9.0	-9.3
Rubber	[45,-15] **86.9**	-2.1	-5.2	-3.1	-2.4
Areca	[30,5] **94.9**	-1.4	-9.3	-4.3	-2.8
Pineapple	[40,-5] **70.4**	-11.6	-17.0	-4.0	-4.2

For the optimal models, the first number in brown brackets [ ] represents the elliptical orientation angle of the GCP model, while the second number represents the ellipticity angle. Black bold numbers in the table indicate optimal models' classification accuracies, and other black numbers show differences in classification accuracy compared to the optimal model.

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The calculations of N_d and $\bar{N}_c$ do not require classification results. They are derived from the inherent statistical properties of the interaction between electromagnetic waves and ground objects, as captured by the Wishart distance. These metrics are designed to evaluate the classification performance of different polarization types in relation to ground objects. The purpose of designing these metrics is to provide a way to identify the optimal polarization for classification without needing to exhaustively test all possible classification methods.

The two parameters proposed in this article allow for identifying the optimal GCP mode for classification simply by calculating the parameter values, without needing to classify all modes. This approach offers significant flexibility in practical use. For instance, when focusing on classifying different tree species or crops, using the overall parameter and the single-class parameter selectively can identify the GCP mode that is optimal overall and best distinguishes specific tree species or crops from others, thereby improving classification performance and providing valuable insights for task design.

In practical applications, our proposed workflow involves using existing fully polarimetric SAR data to design the optimal polarization mode for target categories of interest (e.g., buildings, water ice, or specific vegetation types). Using the proposed metrics, the optimal GCP mode and its corresponding ellipticity angle and orientation angle are determined. From these parameters, the necessary adjustments to the amplitude and phase differences of the H and V channels are calculated using (18) and (19) in Appendix. Based on these results, the SAR system adjusts the transmission settings of the sparse phased array, the activation or deactivation of final-stage power amplifier branches, and the phase differences of the horizontal and vertical channels to achieve optimal elliptical polarization transmission.

It is worth noting that the proposed parameters N_d and $\bar{N}_c$ are designed based on statistical distribution distances, specifically the Wishart distance, and are not related to physical scattering mechanisms. While N_d and $\bar{N}_c$ carry certain interpretative significance, their design is not directly grounded in physical scattering mechanisms.

Currently, research on circular compact polarization is quite popular. However, the results of this article indicate that CP should not be regarded as the "golden configuration" of polarization systems. Instead, different polarization modes should be leveraged according to specific application needs. Quad polarization systems exhibit the best performance in classification tasks due to their ability to obtain complete polarization information. However, their system design is more complex, with heavier payloads that make miniaturization challenging. Additionally, the mapping swath width of quad polarization systems is narrower compared to GCP systems, making frequent revisit capabilities more difficult to achieve. Thus, in future task designs, it is essential to balance these costs and select the most suitable polarization mode according to the actual mission requirements.

It is worth noting that the two optimal modes mentioned above cannot yet be achieved by currently operational SAR systems. Future SAR system research should aim to realize all GCP modes. The average relative Wishart distance difference within the class parameter exhibited oscillations in a few low-accuracy modes within the Building3 and vegetation categories, as well as in the Hainan crop scenes. These categories are primarily characterized by volume scattering, and the main reason for the oscillations is the high complexity of these scenes, which include individual pixels with extremely high parameter values, leading to some discrepancies with the classification results. Future research should aim to mitigate the influence of these particular pixels and explore the potential uses of such unique responses in target detection.

VI. Conclusion

In this study, we first revealed the fundamental equivalence between circular and linear polarization in terms of information content through a comparison of transmit–receive matrices, challenging the assumption that circular polarization inherently provides an advantage in information content. Based on an analysis of typical target scattering matrices, we identified the limitations of traditional linear DP and circular CP modes in certain tasks, emphasizing that the selection of the optimal polarization mode should be task-specific. This innovative perspective challenges the widely accepted assumption of CP superiority and is theoretically validated through experiments.

We further proposed the GCP mode, encompassing all elliptical polarization modes, along with two parameters for selecting and evaluating the optimal polarization mode. Experimental results demonstrate that for certain categories, the GCP mode not only achieves performance comparable to full polarization but also delivers accuracy improvements of up to 2%, 17%, and 42% over CP and DP modes in specific tasks. These improvements are attributed to the interaction between the polarization mode and the classification scene, rather than enhancements in the classification algorithm itself. This finding highlights the potential value of GCP modes in various classification tasks, not only improving classification performance but also providing a scientific basis for the design and mission planning of future polarimetric SAR systems.

We believe these findings hold significant implications for the design, mission planning, and operations of future SAR systems, particularly in providing new methods for task-specific polarization mode selection. Future research could further explore the effects of different frequency bands and resolutions on the selection of GCP modes and extend its application to more complex tasks such as iceberg monitoring, lunar exploration, and deep-space missions. Moreover, the design and calibration of GCP systems merit further investigation, and we hope this study will inspire broader discussions and applications in these areas.

Appendix

A. Elliptical Polarization

The time-domain vector expression of the electric field strength with propagation direction z is

$$\vec{E}(z,t) = \begin{bmatrix} E_{0x}e^{-az}\cos(\omega t - kz + \delta_x) \\ E_{0y}e^{-az}\cos(\omega t - kz + \delta_y) \\ 0 \end{bmatrix}. \tag{19}$$

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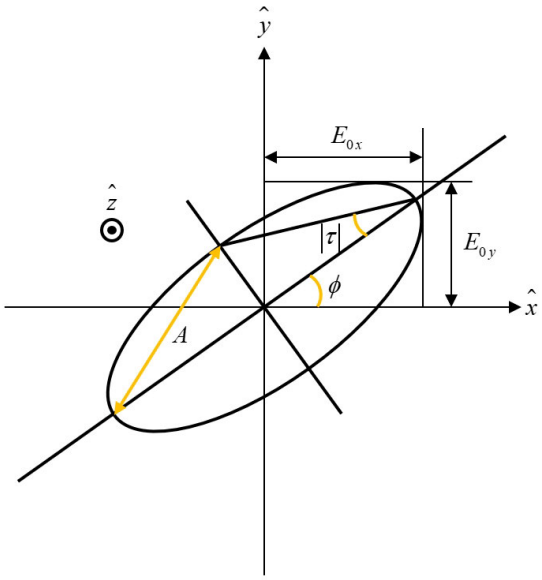


Fig. 15. Polarization ellipse.

The attenuation factor a is the same on all components of the electric field vector and does not affect the polarization characteristics. The amplitude E_{0x} of H is equal to the amplitude E_{0y} of V. The polarization situation is only determined by the phase difference $\delta = \delta_y - \delta_x$. When the phase difference δ is 0, the polarization mode is linear polarization. When the phase difference is $(\pi/2) + n\pi$, the polarization mode is circular polarization. Elliptical polarization refers to all forms of polarization except linear polarization and circular polarization. The polarization ellipse is defined by the following three parameters, as shown in Fig. 15.

1) The amplitude A of the ellipse is determined by the major and minor axes of the ellipse

$$A = \sqrt{E_{0x}^2 + E_{0y}^2}. \tag{20}$$

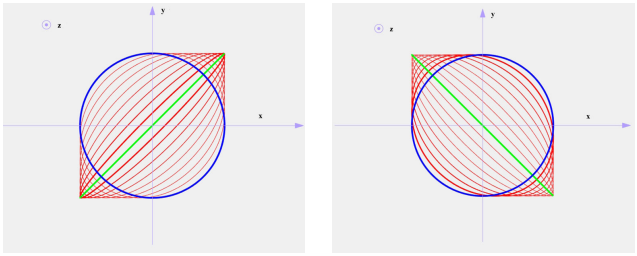
2) The elliptical orientation angle ϕ is defined as the angle between the major axis of the ellipse and the x-axis direction

$$\tan 2\phi = 2\frac{E_{0x}E_{0y}}{E_{0x}^2 - E_{0y}^2}\cos\delta. \tag{21}$$

3) Ellipticity angle τ, defined as

$$|\sin 2\tau| = 2\frac{E_{0x}E_{0y}}{E_{0x}^2 + E_{0y}^2}|\sin\delta|. \tag{22}$$

SAR achieves quad polarization operation by alternating between transmitting H and V polarization. By simultaneously transmitting different amplitudes and phases of H and V, polarization modes of transmitted $\pi/4$, circular polarization, or elliptical polarization are achieved. In current SAR systems, the amplitudes of transmitted H and V polarization are fixed for maximum power transmission, and the ellipticity of the transmitted wave cannot be controlled by adjusting the amplitude. It can only be controlled by changing the phase difference between H and V. Since the amplitude $E_{ox} = E_{0y}$ is fixed and cannot be adjusted, the ellipticity amplitude A of the polarization ellipse is fixed, the ellipticity orientation angle ϕ is $\pi/4$ or $-\pi/4$, the ellipticity angle $\tau = \delta/2$, and the axial ratio is $c\tan(\tau)$. The electromagnetic wave vector $\vec{E}(z_0, t)$ propagating along the z-axis rotates on the (x, y) plane with time to form a polarization ellipse. Fig. 16 shows the change of the polarization ellipse with the change of phase difference.



(a) (b)

Fig. 16. (a) and (b) Change of the polarization ellipse with the phase difference from 0° to 90° and from 90° to 180°, respectively. The polarization changes from linear polarization to elliptical polarization and finally to circular polarization. The x- and y-axes represent the electric field strengths in the x- and y-directions, respectively, and the z-axis represents the propagation direction.

B. Scattering Vectors of DCP Mode and CTLR Mode Under Linear Polarization Basis and Circular Polarization Basis

DCP mode compact polarization transmits either right-hand circularly polarized waves or left-hand circularly polarized waves, and receives both right-hand and left-hand circularly polarized waves. Taking the DCP mode compact polarization SAR system that transmits right-hand circularly polarized waves as an example, the system receives two polarization combination echoes, and the measured scattering vector is

$$S_{\mathrm{DCP}} = \begin{bmatrix} S_{RR} \\ S_{LR} \end{bmatrix} = \frac{1}{2}\begin{pmatrix} -1 & j & j & 1 \\ j & 1 & -1 & j \end{pmatrix}\begin{bmatrix} HH \\ HV \\ VH \\ VV \end{bmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}\begin{bmatrix} LL \\ LR \\ RL \\ RR \end{bmatrix}. \tag{23}$$

CTLR mode compact polarization transmits either right-hand circularly polarized waves or left-hand circularly polarized waves, and receives both H and V linear polarized waves. Taking the CTLR mode compact polarization SAR system that transmits right-hand circularly polarized waves as an example, the system receives two polarization combination echoes, and the measured scattering vector is

$$S_{\mathrm{CTRL}} = \begin{bmatrix} S_{HR} \\ S_{VR} \end{bmatrix} = \frac{\sqrt{2}}{2}\begin{pmatrix} j & 1 & 0 & 0 \\ 0 & 0 & j & 1 \end{pmatrix}\begin{bmatrix} HH \\ HV \\ VH \\ VV \end{bmatrix} = \frac{\sqrt{2}}{2}\begin{pmatrix} 0 & 1 & 0 & -j \\ 0 & -j & 0 & 1 \end{pmatrix}\begin{bmatrix} LL \\ LR \\ RL \\ RR \end{bmatrix}. \tag{24}$$

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C. Scattering Matrices of Typical Targets Under Different Transmitting Bases

In the appendix, S_{QL} represents the scattering matrix when transmitting full-polarization linear polarizations H and V and receiving H and V simultaneously. S_R represents the scattering matrix received in the polarization basis $[h, v]$ when transmitting R polarization, which is essentially $[S_{HR}, S_{VR}]$, simplified as S_R. Other terms like S_H in the appendix follow the same notation.

Trihedral corner reflector

$$S_{QL} = \begin{bmatrix} HH & HV \\ VH & VV \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
$$S_H = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad S_V = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \; S_R = \begin{bmatrix} 1 \\ -j \end{bmatrix}, \; S_L = \begin{bmatrix} 1 \\ j \end{bmatrix}. \quad (25)$$

Dihedral corner reflector

$$S_{QL} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \; S_V = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$
$$S_R = \begin{bmatrix} 1 \\ j \end{bmatrix}, \quad S_L = \begin{bmatrix} 1 \\ -j \end{bmatrix}. \quad (26)$$

Dihedral corner reflector with rotation angle θ

$$S_{QL} = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$$
$$S_H = \begin{bmatrix} \cos 2\theta \\ \sin 2\theta \end{bmatrix}, \quad S_V = \begin{bmatrix} \sin 2\theta \\ -\cos 2\theta \end{bmatrix}$$
$$S_R = \begin{bmatrix} e^{-j2\theta} \\ je^{-j2\theta} \end{bmatrix} = e^{-j2\theta} \begin{bmatrix} 1 \\ j \end{bmatrix}$$
$$S_L = \begin{bmatrix} e^{j2\theta} \\ -je^{j2\theta} \end{bmatrix} = e^{j2\theta} \begin{bmatrix} 1 \\ -j \end{bmatrix}. \quad (27)$$

Horizontal Dipole

$$S_{QL} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad S_H = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \; S_V = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$
$$S_R = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad S_L = \begin{bmatrix} 1 \\ 0 \end{bmatrix}. \quad (28)$$

Vertical Dipole

$$S_{QL} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad S_H = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \; S_V = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$
$$S_R = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad S_L = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (29)$$

Right-handed helix

$$S_{QL} = \frac{e^{-j2\phi}}{2} \begin{bmatrix} 1 & -j \\ -j & -1 \end{bmatrix}, \quad S_H = \begin{bmatrix} 1 \\ -j \end{bmatrix}, \; S_V = \begin{bmatrix} 1 \\ -j \end{bmatrix}$$
$$S_R = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad S_L = \begin{bmatrix} 1 \\ -j \end{bmatrix}. \quad (30)$$

Left-handed helix

$$S_{QL} = \frac{e^{-j2\phi}}{2} \begin{bmatrix} 1 & j \\ j & -1 \end{bmatrix}, \quad S_H = \begin{bmatrix} 1 \\ j \end{bmatrix}, \; S_V = \begin{bmatrix} 1 \\ j \end{bmatrix}$$
$$S_R = \begin{bmatrix} 1 \\ j \end{bmatrix}, \quad S_L = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (31)$$

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